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STATISTICAL TRAFFIC ANALYSIS OF MIX
NETWORKS FOR ANONYMOUS
BROADCASTING IN PRIVACY-PRESERVING
PROOF-OF-STAKE BLOCKCHAINS

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ABSTRACT

In a proof-of-stake blockchain, a lottery weighted by stake selects the validator that publishes the next block. Privacy-preserving protocols conceal the winner cryptographically, yet it must still broadcast its block. An observer of the network can then identify the sender from the timing of its traffic and estimate its stake from the frequency. This dissertation measures that leakage and the protection that an anonymous communication system provides. The complete system is simulated and calibrated to deployed networks. The traffic is protected by mix networks, in which relays delay and reorder messages, and by dummy cover traffic. Two attacks are then run on it, the Bayesian attribution of each block to its sender and the inference of every validator's stake from its sending frequency. By the anonymity trilemma, strong anonymity cannot be obtained together with low latency and low bandwidth, and the question is on which of the two the blockchain can afford to pay. Against attribution, the random delay of the mixes reduces the identification of the sender to a residual just above a random guess, at a small fraction of the delay the protocol tolerates. Stake privacy depends instead on the concentration of stake, a property of the chain rather than of the defence. At the concentration observed in practice the largest stakeholder is identified within weeks. A limit on the stake per validator, supplemented by cover traffic at a quantified bandwidth cost, postpones this to years rather than preventing it. The framework allows a designer to evaluate a defence and to set its parameters under different constraints.

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CHAPTER 1

INTRODUCTION

Encryption protects the content of a message but not the fact of its transmission. On any shared network, a party with access to the underlying links, such as an internet service provider, an internet exchange, or a state-level monitor, can record who transmitted, when, and to whom, even when the content of every message remains encrypted. In the anonymity literature, such a party is modelled as a global passive adversary (GPA), namely an observer who records the origin and timing of every message but neither decrypts nor interferes with it. Anonymous communication (AC) systems are designed to conceal this metadata. Mix networks route messages through relays that delay and reorder them, so that the time at which a message leaves the system no longer identifies the time at which it entered (Chaum, 1981; Piotrowska et al., 2017). They are commonly supplemented with cover traffic, i.e. messages from participants who have nothing to send, which prevents the act of transmission from being informative in itself.

AC systems have historically transported generic user traffic, for example, electronic mail through Chaum’s mixes, web browsing through Tor (Dingledine et al., 2004), and message-based traffic through Loopix and the deployed Nym network (Diaz et al., 2021). Their protection, however, has a cost, as every delay added to a message postpones its delivery and every cover message consumes bandwidth. This trade-off between anonymity, latency, and bandwidth is fundamental (Das et al., 2018). Therefore, the value of an AC system depends on its setting, in particular on how much latency and bandwidth the application can afford and on how much regularity its traffic exposes to the observer.

One setting that is demanding in all three respects is the consensus traffic of blockchains. A blockchain is a public ledger maintained by a decentralised network of computers rather than by a single trusted entity. The network repeatedly selects a participant to append the next block of transactions to the ledger (Nakamoto, 2008; Bonneau et al., 2015). In proof-of-stake (PoS) protocols, this selection is a recurring lottery in which each participant’s probability of winning increases with its stake, i.e. the fraction of the system’s currency that it has committed. Ethereum and Cardano operate on this principle (Kiayias et al.,

2017; Ethereum Foundation, 2026). The lottery is repeated at short, regular intervals called slots. The winner of a slot, referred to as its leader, must broadcast its block to the whole network. Thus, the rate at which a participant produces blocks reflects its stake.

In these mainstream platforms, the identity of each leader is public by design, which exposes the leader to targeted attacks. For example, a leader can be subjected to denial of service at the moment of proposal, bribery, coercion, or legal pressure (David et al., 2018; Boneh et al., 2020). In response, privacy-preserving proof-of-stake (PPoS) protocols conceal the identity of the leader. In Ouroboros Cryptosinous, a prominent example, each participant evaluates the lottery locally with a verifiable random function, and the published block does not name its producer (Kerber et al., 2019). Hence, no other party learns who has won a slot. From a cryptographic standpoint, the leader is hidden.

However, the privacy of the lottery does not extend to the network layer, as the leader must still transmit its block. The block is disseminated to every other participant over a peer-to-peer (P2P) gossip network, and the resulting traffic is visible to the GPA. Even when every payload is encrypted, the GPA learns which participant transmitted first and at what time. Consequently, broadcasts can be linked to their senders, which reveals the leader and thereby restores the targeted attacks. Without an AC system at the network layer, a PPoS blockchain forfeits its privacy guarantee.

Consensus traffic is a particularly difficult and largely untested case for an AC system. The broadcasts occur at fixed slot intervals, their senders are selected in proportion to stake, and every protocol parameter is public. Delay is also more costly here than in most other settings. A block that propagates slowly risks being discarded (Decker and Wattenhofer, 2013), whereas an AC system protects by adding delay (Das et al., 2018). Mainstream platforms have in practice declined such protection. For instance, the Bitcoin proposal of Dandelion++, a protocol designed to anonymise transactions, was rejected (Denby et al., 2017; Fanti et al., 2018).

The Logos Blockchain, under development at the Institute of Free Technology (IFT), implements a PPoS lottery in its consensus protocol, Cryptarchia, and delegates network privacy to the Blend Network, a mix-based service intended to hide the traffic of leaders (Logos, 2026b). Systems of this kind are aimed at users for whom surveillance is a real threat, for example, journalists, activists, and people whose financial activity is monitored (Diaz et al., 2021). However, the protection that such a system offers against an observer who can watch for years, and its cost, must be measured rather than assumed. Existing work leaves this question open. The consensus literature proves leader privacy only by assuming an ideal anonymous broadcast channel (Kerber et al., 2019), and the anonymity literature evaluates its defences on generic traffic (Diaz et al., 2002). This project addresses that gap. It asks whether the traffic reveals the leader even when the cryptography conceals it, how quickly identification accumulates if it does, and what latency and bandwidth an AC system must add to prevent it.

Throughout this dissertation, the system under study is a network of N validators (nodes) running a PPoS consensus over epochs of T slots. Each validator wins each slot independently with a probability that increases with its relative stake, and every block is broadcast through an AC system observed by the GPA. The hypothesis of this study is that the traffic induced by a PPoS blockchain exposes sufficient metadata for the GPA to recover both the identities of leaders and the distribution of stake, and that a mix-based AC system can suppress this leakage to a small and quantifiable residual at a latency and bandwidth cost compatible with the constraints of the consensus protocol. This hypothesis will be tested by constructing a modular simulation of the complete system, comprising the consensus lottery, the gossip network, the AC system, the adversary, and the privacy measures. The simulated traffic of a full epoch is then subjected to two attacks. Exact Bayesian attribution of individual broadcasts, i.e. identifying the sender of an observed block, tests sender unlinkability. Stake inference from observations accumulated across epochs assesses stake privacy. The AC system is studied in two forms, a single-path mix that serves as the baseline and a stratified mix network of the kind used by deployed systems. The two are compared on the same measures of anonymity, latency, and bandwidth.

The hypothesis states a quantifiable residual, and the methodology is built to measure it. The attribution attack is an exact Bayesian posterior over the adversary’s observables. The leakage it reports is thus the most that the attribution of a single block can extract from those observables, and it is not limited by any approximation in the attack. An exact posterior also retains some advantage over chance, and the reported value is never zero. The evaluation therefore reports the amount of leakage rather than its presence. The design question becomes how small the residual can be made and how long an observer must watch before it accumulates into identification.

This dissertation contributes an analytical and computational framework for evaluating AC systems in PPoS settings. It allows a designer to test a candidate defence against a specified adversary and to set the parameters of that defence from measured evidence. The adversary has worst-case knowledge of the system, and the leakage measured against it is thus an upper bound on what a weaker observer could extract. Figure 1.1 illustrates the setting, following one block from the lottery through the AC system to the chain while the GPA observes the traffic.

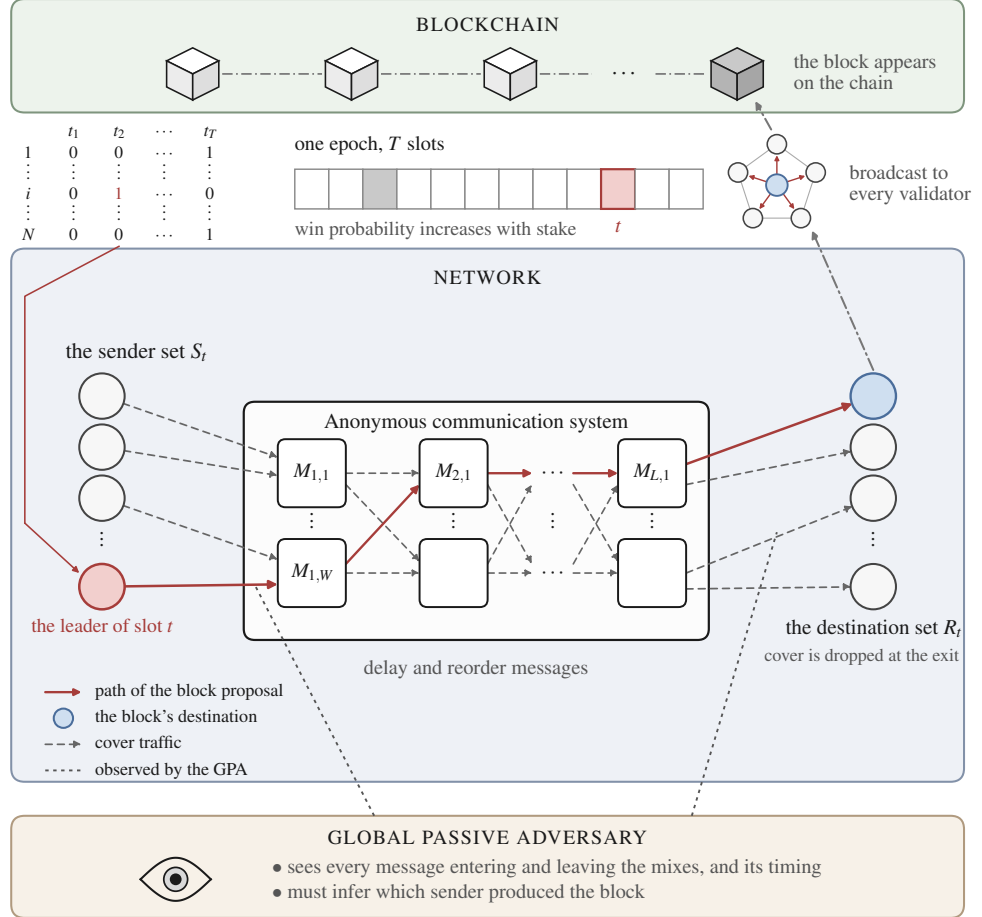


Figure 1.1: The setting of the project. Every validator wins each slot of an epoch independently, with a probability that increases with its stake. The leader of slot t , highlighted in red, must broadcast its block, while a random subset of the other validators transmits cover (dummy) messages in the same slot for obfuscation. Every message in the sender set S_t passes through the AC system, a mix network of L layers of W mixes that delay and reorder what they receive, and then leaves for a random destination. These destinations form the set R_t . The destination of the real block, drawn in blue, broadcasts it onward through the P2P network. The GPA records the origin and timing of every message entering and leaving the mixes.

CHAPTER 2

LITERATURE REVIEW

This chapter reviews the four areas of work on which the dissertation is built, namely AC systems, traffic analysis, the measurement of anonymity and its cost, and PoS consensus with private leader election.

2.1 ANONYMOUS COMMUNICATION SYSTEMS

The modern study of AC systems starts with Chaum (1981), who introduces the mix. A mix is a relay that collects encrypted messages, delays them, and releases them in a permuted order, so that an observer cannot match an output to its input. Later designs remove the trust placed in any single relay by routing each message through a sequence of mixes. However, a mix cannot release a message until enough other messages have arrived to conceal it (batching), which makes its delay long and unpredictable. This is acceptable for electronic mail but unsuitable for interactive applications. In contrast, onion routing (Goldschlag et al., 1996) prioritises low latency and wraps each message in layered encryption, from which every relay learns only its adjacent hops. Its successor, Tor, is the most widely deployed AC system today (Dingledine et al., 2004). Such relays do not delay or reorder messages, and Tor is therefore unable to resist a GPA. The two ends of the design space are reconciled by continuous-time mixing, introduced in the stop-and-go mix of Kesdogan et al. (1998). Batching is replaced by an exponentially distributed delay drawn independently for every message, and the mean delay thereby becomes a tunable parameter rather than a consequence of load. Loopix (Piotrowska et al., 2017) develops this principle into the current state of the art, sustaining mixing with cover traffic when genuine traffic is sparse and arranging the relays in a stratified topology. Thus, strong anonymity is retained at moderate latency, and the residual cost is borne in bandwidth. Its design is deployed at scale by the Nym network (Diaz et al., 2021).

The blockchain setting has produced AC designs of its own because every transaction is broadcast over an observable P2P network. Early measurement studies demonstrate

that an observer of the Bitcoin network can link transactions to the internet protocol (IP) addresses that originate them (Koshy et al., 2014; Biryukov et al., 2014). However, reusing a general-purpose system is not a solution, as routing Bitcoin traffic through Tor introduces new attacks (Biryukov and Pustogarov, 2015). Defences built into the blockchain protocol itself follow. Dandelion++ (Fanti et al., 2018) anonymises the origin of a transaction by sending it along a random relay path before diffusing it to the whole network. Its Bitcoin proposal was rejected over denial-of-service concerns (Denby et al., 2017), although the privacy-focused cryptocurrency Monero later adopted it (Sharma et al., 2023). The same study finds that the anonymity methods deployed by cryptocurrencies fall well short of their formal guarantees. Thus, under a GPA, mix networks remain the credible defence, and Nym already transports cryptocurrency traffic (Diaz et al., 2021). These designs hide the origin of transactions, whereas this project must hide the origin of the block itself. The PPoS setting complicates the problem considerably. Block traffic is sparse because most slots elect no leader, and cover traffic is therefore a precondition for anonymity. The defences evaluated in this dissertation are thus mix-based and supplemented with cover. Specifically, the baseline is a single-path chain of mixes with exponential delays, and the primary defence is a stratified mix network of the Loopix kind, characterised by its width and depth.

2.2 TRAFFIC ANALYSIS

There are two kinds of passive attack on AC systems. The first relies on a single observation, for example, the timing of one message as it crosses the network. The second accumulates observations over a long horizon. Danezis (2003) formalises the latter as the statistical disclosure attack (SDA). An observer who monitors many rounds of communication identifies the persistent recipients of one sender, because the traffic of all other senders averages out. Mathewson and Dingledine (2004) extend this attack to pool mixes, partial observation, and padded traffic, and conclude that against a persistent pattern the attack eventually succeeds. Hence, the effectiveness of a defence is measured by the number of observations the adversary needs, rather than by whether it prevents success. Troncoso and Danezis (2009) then reformulate traffic analysis as Bayesian inference, treating the unobserved routes through the network as latent variables and computing the posterior probability of each candidate sender given the observations. For a given inference on a fixed set of observables, the exact posterior is the strongest attack available. On the defensive side, Oya et al. (2014) quantify the limits of dummy, i.e. cover, traffic against SDAs and show how a dummy strategy can be designed towards a stated privacy target. The slot-synchronous cover traffic studied in this project is a sender-based dummy strategy in this sense, and its rate is likewise set against such a target. This dissertation adopts the Bayesian framing, under which the measured leakage is the most that the observables al-

low. Its stake-inference attack is a stake-weighted counterpart of the long-term disclosure attack, applied to the recurring lottery of a PPoS blockchain.

2.3 MEASURING ANONYMITY AND ITS COST

Evaluating an attack requires a measure of anonymity. The earliest such measure is the size of the anonymity set, i.e. the number of candidates that could have sent the observed message, each treated as equally likely. Serjantov and Danezis (2002) and Diaz et al. (2002) show that this count is misleading once the candidates differ in probability. Both instead propose the entropy of the adversary’s posterior distribution, from which an effective anonymity-set size follows. These measures are adopted here because the timing evidence makes the candidates unequal, and the size of the set would overstate the protection. Notably, Diaz et al. (2002) also leave the evolution of anonymity under prolonged observation and the quantitative benefit of cover traffic as open problems, both of which are addressed in this project.

Das et al. (2018) quantify the cost of anonymity by proving the anonymity trilemma, which states that strong anonymity cannot be achieved together with both low latency overhead and low bandwidth overhead. This result establishes the three axes on which any defence must be evaluated. More recently, Das et al. (2024) provide the first formal analysis of continuous mix networks of the Loopix kind and derive a lower bound on the adversary’s advantage that holds even when every relay is honest. Hence, some residual leakage is unavoidable, and the question is its size.

For stake privacy, the literature provides no comparable measure, perhaps because stake does not arise in generic traffic. The closest concept is the sending rate of a user, which in a PPoS blockchain reflects stake. This dissertation therefore extends the same principle and measures whether the adversary identifies the largest stakeholder, and the group of the largest stakeholders, from their sending frequencies. The entropy and posterior measures are applied throughout, and the defence configurations are reported with their position on the trilemma axes.

2.4 PROOF OF STAKE AND PRIVATE LEADER ELECTION

In proof-of-work, the right to append a block is earned by computation (Nakamoto, 2008). PoS replaces this computation with a lottery weighted by stake. Ouroboros (Kiayias et al., 2017) establishes the first provably secure PoS protocol, and its successor, Ouroboros Praos (David et al., 2018), introduces the private per-slot lottery adopted in this project. Every validator evaluates a verifiable random function locally and wins each slot independently with probability $\phi(\alpha) = 1 - (1 - f)^\alpha$, where $f \in (0, 1)$ is the active slots coefficient and α is the relative stake of the validator. This probability increases with stake. Because the

relative stakes sum to one, f is also the probability that a slot has at least one leader. On the network side, Decker and Wattenhofer (2013) measure block propagation in Bitcoin and show that a block that propagates slowly risks exclusion from the chain. This finding bounds the delay that an AC system may add.

A separate line of research addresses privacy in blockchains. Zerocash (Ben-Sasson et al., 2014) hides the payments recorded on a ledger, and Ouroboros Cryptosinous (Kerber et al., 2019) extends this protection to the consensus itself. A Praos block identifies its producer, whereas a Cryptosinous block proves its validity without identifying the validator that created it. However, the Cryptosinous security proof assumes an ideal anonymous broadcast channel, which places the network layer outside the model. The Logos Blockchain brings this line of work into practice. Its PPoS protocol, Cryptarchia, delegates the broadcast of blocks to the Blend Network, a mix-based service of the kind reviewed above (Logos, 2026b).

2.5 THE GAP

The consensus literature and the AC literature have not been connected. The consensus proofs assume the anonymous channel that the AC systems could supply, and the AC systems are evaluated on traffic unlike the one they would have to protect. Consensus traffic is not generic. Its timing, its stake-weighted senders, and its protocol parameters are all known to the adversary. The weighting by stake also raises a question that no generic AC study asks, in particular whether the distribution of stake itself can be inferred from traffic. A PPoS ledger conceals stake as part of its privacy goal, and a network layer that exposes it would reveal, through traffic, what the cryptography conceals on the chain. What remains missing is a measurement of how much protection an AC system of the deployed kind provides for a PPoS blockchain under a GPA, at what latency and bandwidth cost, and for how long that protection lasts under sustained observation. This measurement is the purpose of this dissertation.

CHAPTER 3

SYSTEM AND THREAT MODEL

This chapter specifies the system whose traffic is analysed and the adversary that observes it. Figure 3.1 shows how its components compose into one pipeline. The consensus lottery determines which validators send and in which slots. The network model assigns each message its propagation latency. The AC system delays and reorders the resulting traffic to obscure the timing that would otherwise reveal the sender, and it emits the record of observable events, referred to as the trace. From the trace, the GPA computes a guess, either a posterior over the candidate senders of a block proposal or an estimate of every validator’s stake. The measures evaluate each guess against the ground truth.

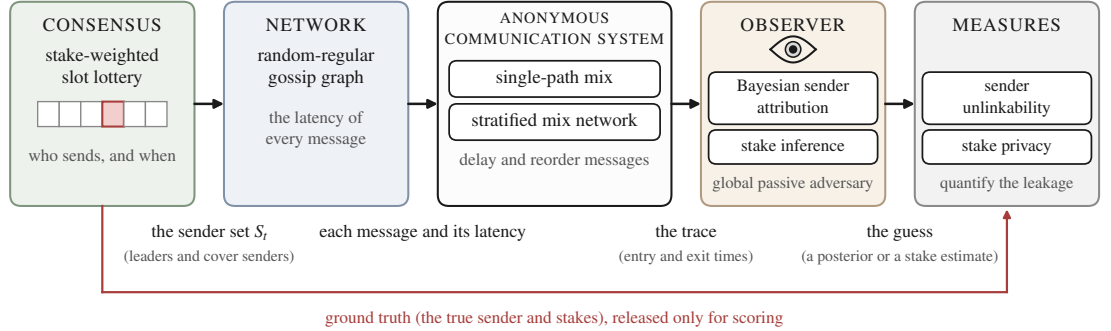


Figure 3.1: The evaluation pipeline. The sender set S_t of a slot comprises its leaders, if any, and the cover senders. Either of the two AC systems shown can be substituted without changing the rest of the pipeline. The trace records the entry and exit time of every message and is all that the GPA observes. The ground truth, drawn in red, is withheld from the GPA and released only for scoring.

3.1 CONSENSUS TRAFFIC

Consensus traffic is modelled as the output of the leader election among N validators. A validator i holds a stake $\omega_i > 0$, and its relative stake is $\alpha_i = \omega_i / \sum_{j=1}^N \omega_j$, so that every $\alpha_i \in (0, 1)$ and $\sum_{i=1}^N \alpha_i = 1$. In each slot, validator i wins the election with the probability

given by the lottery function:

$$\phi(\alpha_i) = 1 - (1 - f)^{\alpha_i}, \quad (3.1)$$

where $f \in (0, 1)$ is the active slots coefficient (David et al., 2018; Kerber et al., 2019). The function is strictly increasing in the stake, with $\phi(0) = 0$ and $\phi(1) = f$, and hence $0 < \phi(\alpha_i) < f$ for every validator. The outcome of the lottery is private, i.e. a win is known only to the winning validator until the corresponding block is broadcast (Kerber et al., 2019). The cryptographic mechanism that provides this privacy is not simulated. Its effect is represented by sampling the outcomes directly and withholding them from the GPA.

Time is divided into epochs of T slots, labelled $t = 1, \dots, T$, and every validator participates in the election in every slot. The outcome for validator i in slot t is the binary variable $s_i(t) \in \{0, 1\}$, equal to 1 if the validator wins the slot. Wins are independent across validators and across slots. Thus, the outcome $\mathbf{s}(t) = (s_1(t), \dots, s_N(t))$ of a slot follows the product law:

$$\mathbb{P}(\mathbf{s}(t)) = \prod_{i=1}^N \phi(\alpha_i)^{s_i(t)} [1 - \phi(\alpha_i)]^{1-s_i(t)}. \quad (3.2)$$

The simulation samples this law slot by slot and records the win events, specifically the pairs (t, i) with $s_i(t) = 1$. Each event is one block proposal that must be broadcast, and this stream of events is the consensus traffic processed by the rest of the pipeline. Two constants fix the scale, with values representative of the Logos Blockchain (Logos, 2026a,b). The active slots coefficient is the Cryptarchia protocol constant $f = 1/30$, and with slots of one second, a block is produced every 30 s on average. The epoch length is $T = 388,800$ slots, i.e. 4.5 days, the window over which the Logos protocol observes chain growth to infer the total active stake.

The traffic that reaches the network is governed by the number of leaders in a slot, $n_t = \sum_{i=1}^N s_i(t)$. Under equation (3.2), n_t is a sum of independent Bernoulli variables with unequal success probabilities and therefore follows a Poisson binomial distribution, with moments (see Appendix A.1):

$$\mathbb{E}[n_t] = \sum_{i=1}^N \phi(\alpha_i), \quad \text{Var}(n_t) = \sum_{i=1}^N \phi(\alpha_i) [1 - \phi(\alpha_i)], \quad (3.3)$$

and the average number of block proposals per epoch is $T \sum_{i=1}^N \phi(\alpha_i)$.

For small stakes, the mean expands:

$$\sum_{i=1}^N \phi(\alpha_i) = -\ln(1 - f) - \frac{\ln^2(1 - f)}{2} \sum_{i=1}^N \alpha_i^2 + O\left(\sum_{i=1}^N \alpha_i^3\right), \quad (3.4)$$

whose leading term $\lambda = -\ln(1 - f)$ does not depend on the stake vector. The stake distribution therefore enters the mean only at second order, through the concentration term $\sum_{i=1}^N \alpha_i^2$. Moreover, because every $\phi(\alpha_i) \leq f$, equation (3.3) implies $(1 - f) \mathbb{E}[n_t] \leq \text{Var}(n_t) \leq \mathbb{E}[n_t]$, and n_t is well approximated by a Poisson variable with rate λ .

The distribution of n_t also has an exact property. The probability that a slot has at least one leader is the active slots coefficient itself:

$$\begin{aligned} \mathbb{P}(n_t \geq 1) &= 1 - \prod_{i=1}^N [1 - \phi(\alpha_i)] \\ &= 1 - (1 - f)^{\sum_{i=1}^N \alpha_i} \\ &= f, \end{aligned} \tag{3.5}$$

where the final step uses $\sum_{i=1}^N \alpha_i = 1$. Notably, this holds for every N and every admissible stake vector. Thus, an epoch contains $Tf = 12,960$ slots with at least one leader on average, fewer than the expected number of proposals because a slot with several leaders yields several proposals. The probability of such slots is (see Appendix A.1):

$$\mathbb{P}(n_t \geq 2) = f - (1 - f) \sum_{i=1}^N \frac{\phi(\alpha_i)}{1 - \phi(\alpha_i)}. \tag{3.6}$$

Figure 3.2 compares the distribution of n_t over one simulated epoch with the Poisson law of rate λ . Theory and simulation agree at every leader count within sampling error.

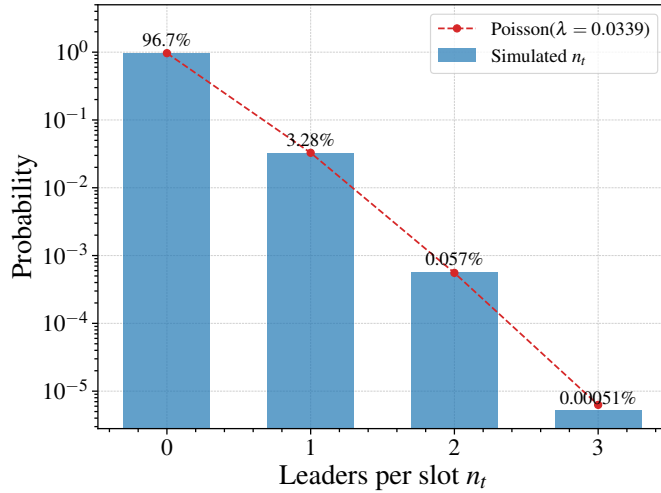


Figure 3.2: The distribution of the number of leaders per slot over one simulated epoch ($N = 1000$, $T = 388,800$, $f = 1/30$) against the Poisson law with rate $\lambda \approx 0.0339$. Each bar is annotated with the simulated fraction of slots with that number of leaders.

These proportions determine the structure of the induced traffic. In nearly all slots, no block proposal is transmitted, and a slot with at least one proposal almost always has exactly one. Consequently, an unprotected broadcast would appear as a single transmission

in an otherwise silent network, and the GPA would identify its sender easily. Furthermore, n_t is a random variable rather than a constant, and so is the size of any candidate set constructed around an observed broadcast.

The form of equation (3.1) is the only one that keeps the probability that a slot has a leader independent of the stake vector. If a lottery function is required to satisfy $\mathbb{P}(n_t \geq 1) = f$ for every N and every admissible stake vector, then $h(x) = 1 - \phi(x)$ must satisfy $\prod_{i=1}^N h(\alpha_i) = 1 - f$. Taking logarithms turns this condition into Cauchy's functional equation, whose bounded solution is $h(x) = (1 - f)^x$ (Aczél and Oser, 2006). The same product form yields $[1 - \phi(\alpha)][1 - \phi(\alpha')] = 1 - \phi(\alpha + \alpha')$, and hence the probability that at least one member of a group of validators wins a slot depends only on their aggregate stake and is unaffected by splitting or pooling (David et al., 2018). Consequently, an observer who counts the occupied slots learns only f , which is public. The block count is only marginally more informative, as its mean depends on the stake vector through the second-order term of equation (3.4), which lies below the sampling noise of one epoch. Anything that the GPA learns about stake must therefore come from attributing individual broadcasts to their senders. In contrast, the win rate of a single validator can be inverted through the strictly increasing ϕ into an estimate of its stake, and this inversion is the basis of the stake inference attack.

The final component of the consensus model is the stake distribution. Stake in deployed PoS networks is heavily concentrated, with a few large stakeholders and a long tail of small ones. The raw stakes are therefore drawn from a heavy-tailed distribution, specifically the Pareto distribution of type I, whose density is:

$$\frac{k x_m^k}{\omega^{k+1}} \quad \text{for } \omega \geq x_m, \quad (3.7)$$

where $k > 0$ is the shape and $x_m > 0$ the scale (Arnold, 2015). Each raw stake ω_i is sampled independently from this density, and the relative stakes follow from the normalisation $\alpha_i = \omega_i / \sum_{j=1}^N \omega_j$, in which the scale cancels. The scale is thus fixed at $x_m = 1$, and the shape k is the only free parameter. A small k produces a heavy tail in which a few validators hold most of the stake, whereas a large k spreads the stake almost evenly, with every share close to $1/N$. The Pareto distribution is chosen over other heavy-tailed candidates as the classical one-parameter model of wealth concentration, whose inequality has a closed form through which k can be calibrated (Arnold, 2015). The model can thus be matched to the stake distribution of a real blockchain through a single inequality statistic.

For the Pareto distribution, the Gini coefficient $G \in (0, 1)$ has the closed form (Arnold, 2015):

$$G = \frac{1}{2k - 1}, \quad \text{equivalently} \quad k = \frac{1 + G}{2G}. \quad (3.8)$$

The model is calibrated to Cardano, the largest deployed network whose leader election

uses the lottery of equation (3.1). The relevant population is the stake pools rather than the individual token holders, because pools win blocks in proportion to their active stake and thus play the role of the validators. According to Messari’s State of Cardano report, the top 6% of pools hold half of the active stake (Messari, 2024). This top-share statistic is converted into k through the Lorenz curve of the Pareto distribution, i.e. the cumulative share of stake held by the poorest fraction u of validators (Arnold, 2015):

$$\Lambda(u) = 1 - (1 - u)^{(k-1)/k}. \quad (3.9)$$

The share held by the richest fraction u is the complement $1 - \Lambda(1 - u) = u^{(k-1)/k}$, and solving $u^{(k-1)/k} = 1/2$ at $u = 0.06$ yields $k \approx 1.33$, i.e. $G \approx 0.60$. A direct measurement of the pool active-stake vector through the Koios API confirms this statistic, with the top 5.9% of pools holding half of the stake at epoch 638 (Koios, 2026). The shape of the Pareto stake distribution is therefore $k = 4/3$.

Cardano’s pool distribution is not exactly Pareto. The same measurement yields a pool-level Gini of 0.84, which would imply $k \approx 1.10$ by equation (3.8), whereas $k = 1.33$ follows from the top-share statistic. Figure 3.3 overlays the measured vector on the calibrated model at the same pool count $N = 2706$. The model places its pools in a narrow band around the uniform share $1/N$ with a single tail towards the large shares. In contrast, the Cardano pools form two populations, a group of large pools whose shares end below the largest shares of the model, and a long flat tail of small pools far below anything the model produces. Hence, the calibration matches the top-share statistic rather than the shape of the whole distribution.

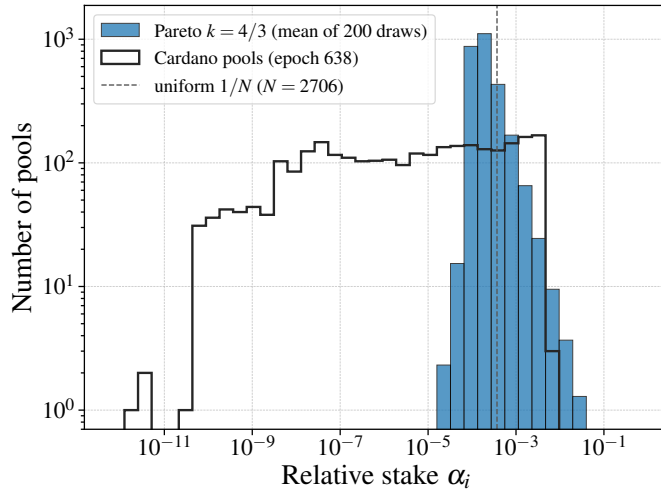


Figure 3.3: Histogram of relative stake for the Cardano pools at epoch 638 ($N = 2706$, Koios) and for the Pareto model with $k = 4/3$ at the same N (mean over 200 draws).

This gap suggests that a mixture of two Pareto components, perhaps one for the large pools and one for the tail of small ones, could capture both statistics. However, a single

shape is assumed here, and the top-share calibration is preferred for two reasons. The top of the distribution is what matters for the questions studied, because high-stake validators win and broadcast most often. In addition, $k \approx 1.10$ lies in the region where finite samples cease to be faithful.

Figure 3.4 examines this faithfulness by comparing the Gini of sampled stake vectors, computed with the standard empirical estimator (Dixon et al., 1987), with the closed form across $k \in [1, 10]$. For $k \geq 1.5$ the sampled Gini tracks $1/(2k - 1)$ to within a few per cent. Near $k = 1$, the Pareto mean diverges and the draw no longer self-averages. Consequently, the usable inequality range at the simulated size $N = 1000$ extends to roughly $G = 0.8$, and the baseline $k = 4/3$ lies within it.

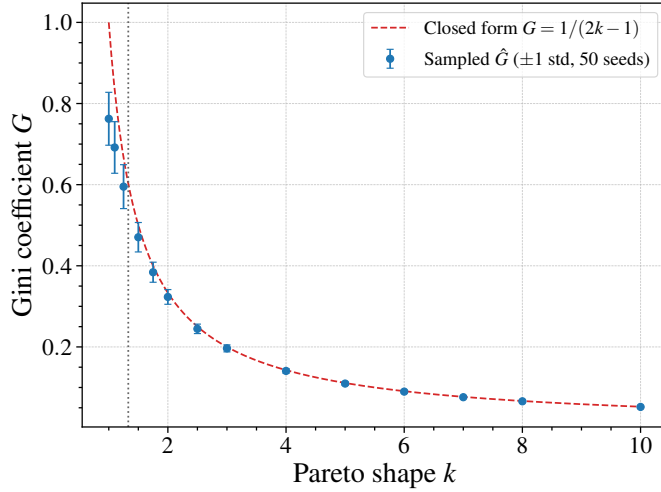


Figure 3.4: The Gini coefficient of sampled stake vectors ($N = 1000$, mean $\pm$ one standard deviation over 50 seeds) against the closed form $G = 1/(2k - 1)$. The dotted vertical line marks the Cardano-calibrated baseline $k = 4/3$.

3.2 NETWORK AND LATENCY

The network model simulates the P2P gossip network and its latencies. These comprise the global broadcast latency, i.e. the time for a block released by the AC system to reach every validator, and the latencies of the AC path.

The gossip network is modelled as a random C -regular graph (RRG) on the N validators, in which every validator is connected to exactly C peers chosen at random. The connectivity is set to $C = 8$, the recommended peering degree of the Logos gossip layer (Logos, 2026b). RRGs are an idealised model of P2P networks. They are sparse, every node has the same degree, and the distance between two randomly chosen nodes increases only logarithmically with N (Bollobás and Fernandez De La Vega, 1982). The graph is sampled once per experiment realisation and held fixed thereafter, as is the stake vector. Quantities frozen in this way are referred to as quenched.

Each edge e of the graph has its own latency d_e , drawn from the link-latency law specified below. A block propagates by gossip along the paths of least total latency, and its broadcast is complete when it reaches the validator farthest from its source in time. The expected global broadcast latency is therefore:

$$\mathbb{E}[D_{\text{br}}] = \mathbb{E} \left[\max_{v \neq u} \min_{\xi: u \rightarrow v} \sum_{e \in \xi} d_e \right], \quad (3.10)$$

where u is the source and ξ ranges over the paths from u to validator v . The inner minimum is the least total latency from u to v , and the outer maximum is the latency to the farthest validator, i.e. the weighted eccentricity of u . This quantity has no closed form and is computed with Dijkstra's shortest-path algorithm over the drawn edge latencies (Dijkstra, 2022). Since the graph distances grow only logarithmically in N (Bollobás and Fernandez De La Vega, 1982), $\mathbb{E}[D_{\text{br}}] = O(\log N)$, and the broadcast latency increases only weakly with the size of the network.

The broadcast is the last stage in the delivery of a block proposal, which first traverses the AC path. One law assigns the latency of every link, on the AC path and the gossip graph alike. The latency of link e for message m is:

$$d_e(m) = d_0 + X_e + \varepsilon_{e,m}, \quad \text{where} \quad \ln X_e \sim \mathcal{N}(a, b^2) \quad \text{and} \quad \varepsilon_{e,m} \sim \text{Exp}(\lambda_\varepsilon). \quad (3.11)$$

The floor $d_0 \geq 0$ is a hard lower bound on the latency of any link, X_e is the structural component of the link, and $\varepsilon_{e,m}$ is a per-message congestion jitter. The structural component is quenched, as are the graph and the stake vector, because it represents the part of latency fixed by geography and routing, which changes slowly relative to an epoch. In contrast, the jitter is redrawn for every message, as queueing varies from message to message. Quantities resampled at every event are referred to as annealed.

The log-normal law of the structural component is chosen for several reasons. Firstly, an end-to-end latency arises from many per-segment effects that compose multiplicatively, and the log-normal is the natural limit law. Secondly, it is continuous and strictly positive, whereas a normal distribution allows negative latencies. Thirdly, its two parameters map one-to-one onto the mean and variance of X_e through:

$$b^2 = \ln \left(1 + \frac{\text{Var}(X_e)}{\mathbb{E}[X_e]^2} \right), \quad a = \ln \mathbb{E}[X_e] - \frac{b^2}{2}. \quad (3.12)$$

Hence, the spread of the link latencies is set directly. The choice is also the least restrictive one, as the log-normal is the maximum entropy distribution of a positive variable whose logarithm has a given mean and variance (Park and Bera, 2009). The floor d_0 prevents draws unrealistically close to zero and, being added to every draw, raises the mean while leaving the variance unchanged.

The constants d_0 , a , and b are calibrated to measured internet latencies from the WonderNetwork ping database (WonderNetwork, 2026). The data set covers 12 cities, four in each of Europe, North America, and Asia, including both major interconnection hubs and less central cities, so that the measured latencies span realistic routes within and between continents. Matching the minimum, mean, and standard deviation of the halved, continent-weighted round-trip times yields $d_0 = 7.2$ ms, a link mean of 62.9 ms, and a link standard deviation of 33.3 ms, i.e. $(a, b) = (-3.041, 0.553)$ with X_e measured in seconds.

The parameters are matched to the moments rather than fitted on the logarithmic scale. The ping population mixes intra-continental and inter-continental routes, and a fit on the logarithms absorbs the separation between the two into the spread parameter, returning 58.6 ms against the measured 33.3 ms, whereas moment matching reproduces the measured value by construction.

Under this calibrated law, equation (3.10) is evaluated by simulation. The quenched graph is sampled, every edge draws its latency from the law, and the weighted eccentricity is averaged over sources and graph realisations. Figure 3.5 shows the result across network sizes. At $N = 1000$ and $C = 8$ the broadcast latency is $\mathbb{E}[D_{\text{br}}] \approx 0.25$ s.

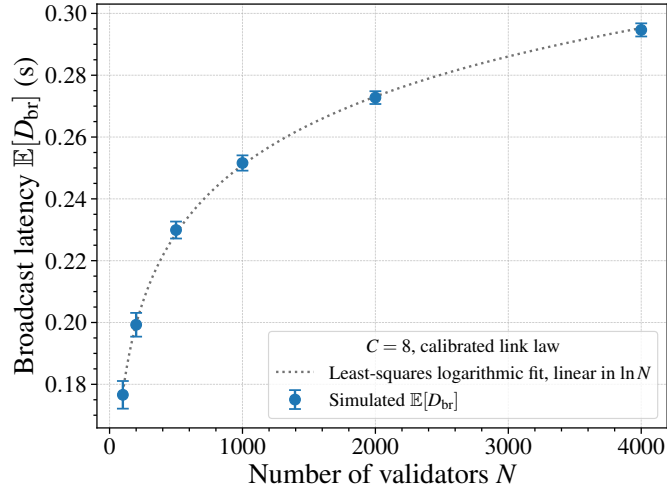


Figure 3.5: The simulated global broadcast latency under the calibrated link law without jitter ($C = 8$, mean $\pm$ one standard deviation over 100 quenched graphs with 64 random sources each). The dotted line is a least-squares fit, linear in $\ln N$, consistent with the $O(\log N)$ growth of the graph distances.

The remaining parameter of the link law is the jitter rate λ_ϵ . The exponential distribution has a single parameter, so its mean $1/\lambda_\epsilon$ equals its standard deviation, and per-message variability cannot be introduced without an equal increase in mean delay. Over a path of several links, the total jitter is a sum of independent exponential draws, one per link, and follows a Gamma distribution. The magnitude of the jitter is treated as an experimental variable, as the aim is to measure how per-message variability affects attribution rather than to reproduce the congestion of any particular network.

Finally, the AC system is external to the gossip network, i.e. its mix nodes are dedicated machines outside the validator set, and the links of the AC path are drawn from the link law rather than routed through the gossip graph. This choice reflects the Logos design, in which the Blend Network is an opt-in overlay service separate from the base gossip protocol (Logos, 2026b). The internal alternative, in which the mixes are themselves validators and every AC-path link is a shortest-path distance through the gossip graph, is not modelled.

3.3 THE SINGLE-PATH MIX

In a PPoS blockchain, the identity of the leader is intended to remain hidden. A block reveals the slot in which it was created but not the validator that produced it, and the GPA therefore attempts to correlate the traffic it observes with the appearance of each block on the chain. A natural first defence, shown in Figure 3.6, routes the block proposal through an onion-routed path of relays before it is broadcast, similar to Tor (Dingledine et al., 2004). In the present setting, most slots produce no proposal, and when the leader transmits after winning a slot, the GPA correlates the only transmission in the slot with the block that appears on the chain. The sender is identified by the mere fact of having transmitted, and a Tor-like path thus provides little protection.

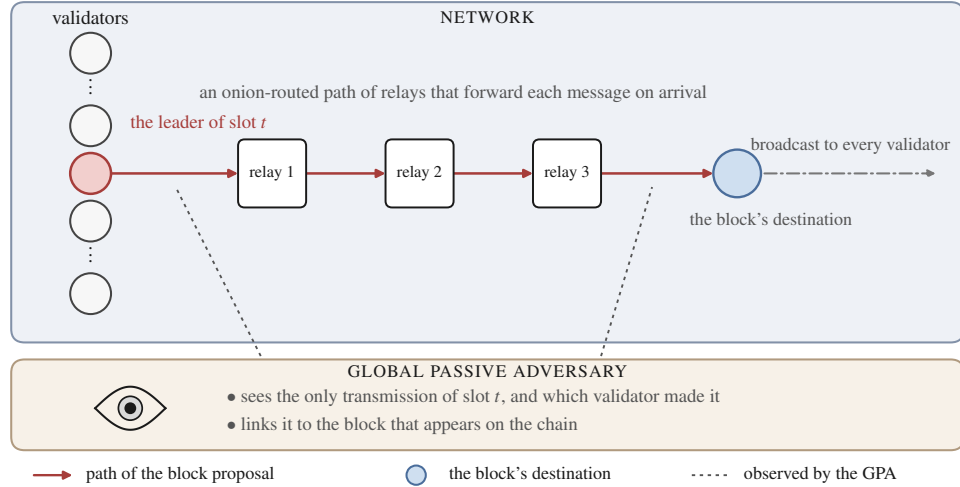


Figure 3.6: A Tor-like approach. Only the leader transmits, along a path of relays that forward each message on arrival. Its transmission is the only one in its slot. Hence, the GPA links it to the block that subsequently appears on the chain.

The first improvement is to introduce cover traffic. In every slot, whether or not a leader exists, a subset of validators sends a message, which contains the genuine block proposal if the validator is the leader and is a cover message otherwise. The act of transmission is then no longer informative by itself. However, cover on its own is insufficient. If every sender routes its message along its own path, as in Figure 3.7, and every relay forwards each message on arrival, the timing of each message is preserved along its path.

Consequently, the GPA can match every exit to its entry and identify the path that delivered the block. Cover must therefore be combined with a mechanism that makes the messages of different senders indistinguishable while they traverse the network.

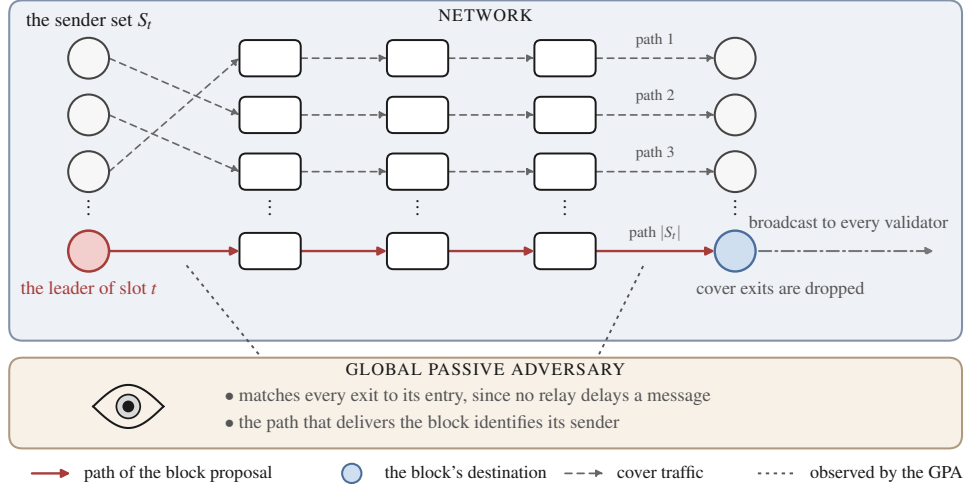


Figure 3.7: Cover traffic over independent paths. The cover senders transmit in the same slot as the leader, but each of the $|S_t|$ senders uses its own path of relays, and no relay delays a message. The timing along each path is preserved. Thus, the GPA identifies the path that delivered the block.

These two requirements produce the single-path mix, shown in Figure 3.8. In every slot t , the senders of the slot, forming the set S_t , inject their messages into one shared path of L mix nodes:

$$S_t \rightarrow M_1 \rightarrow M_2 \rightarrow \dots \rightarrow M_L \rightarrow R_t. \quad (3.13)$$

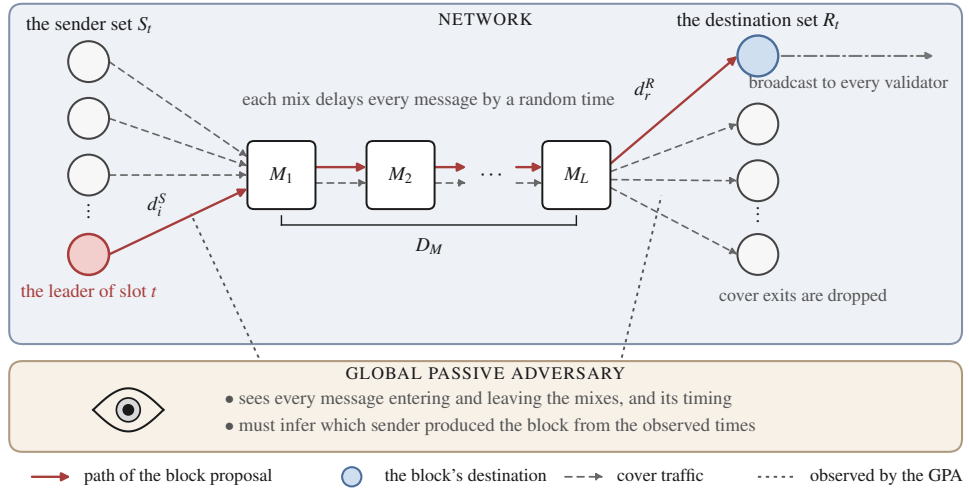


Figure 3.8: The single-path mix. Every sender of slot t injects its message into one shared path of L mixes. The deterministic latencies d_i^S , D_M , and d_r^R are marked.

Each mix delays every message for an independent random time before forwarding it, which breaks the correspondence between the order of entry and the order of exit

(Chaum, 1981; Kesdogan et al., 1998). Each message then leaves the last mix for its own random destination among the N validators, and these destinations form the set R_t . The destination of the real message broadcasts the block through the gossip network, cover messages are discarded on arrival, and the destination adds no delay.

Although highly idealised, this model retains the essential components of an AC system under a GPA, specifically multiple simultaneous senders, cover traffic, shared mixing, and randomised forwarding. It is mathematically tractable and a natural intermediate step towards more realistic AC systems such as the stratified mix network. It is therefore adopted as the baseline defence of this dissertation.

The cover traffic is slot-synchronous. Each validator that has not won a slot emits a cover message in that slot independently with probability p_s , and every message is encrypted and of a common size. Hence, messages differ only in their endpoints and timing. The sender set of slot t comprises the n_t leaders together with the cover senders, and its size is:

$$|S_t| = n_t + B_t, \quad B_t \sim \text{Binomial}(N - n_t, p_s). \quad (3.14)$$

Approximately $p_s N$ validators transmit in each slot. A validator i belongs to S_t either because it wins the slot, with probability $\phi(\alpha_i)$, or because it does not win but sends a cover message. Its participation probability is:

$$q_i = \mathbb{P}(i \in S_t) = p_s + (1 - p_s) \phi(\alpha_i), \quad (3.15)$$

which increases with stake. Cover therefore dilutes the dependence of a validator's sending frequency on its stake but does not remove it. Moreover, $|S_t|$ varies from slot to slot, and any reference quantity computed from it, for example, the uniform guessing probability $1/|S_t|$, is an expectation over the law of equation (3.14) rather than a single number. Where an experiment requires a candidate set of near-deterministic size, a control variant replaces the per-validator rule with a fixed number of cover senders per slot.

The timing of a message combines the latencies of the links it traverses with the intentional random delays introduced by the AC system. The path from sender i to destination r comprises $L + 1$ links, and the quenched components of their latencies sum to the deterministic path latency:

$$\mu(i, r) = d_i^S + D_M + d_r^R, \quad (3.16)$$

where d_i^S is the quenched latency of the link from sender i to the first mix, D_M the total quenched latency of the $L - 1$ links between consecutive mixes, and d_r^R the quenched latency of the link from the last mix to destination r . Since the mix chain is shared, D_M is common to all messages, whereas d_i^S and d_r^R depend on the sender and the destination. The per-message jitters of the $L + 1$ links, denoted $\varepsilon_1, \dots, \varepsilon_{L+1}$, are redrawn for every

message and add to $\mu(i, r)$.

The intentional random delays are exponential (Kesdogan et al., 1998; Das et al., 2024). The sender delays its message by X_S before it enters the path, which spreads the entries of a slot's senders in time, and each mix j then delays it by an independent $X_{M,j}$. The total intentional random delay is:

$$Z = X_S + \sum_{j=1}^L X_{M,j}, \quad \text{where } X_S \sim \text{Exp}(\lambda_S) \quad \text{and} \quad X_{M,j} \sim \text{Exp}(\lambda_M) \quad \text{for } j = 1, \dots, L, \quad (3.17)$$

and the sum of the L mixing delays follows a Gamma distribution with shape L . Three constants parameterise these delays. The depth defaults to $L = 3$, the convention of deployed onion-routing and mix-network designs (Dingledine et al., 2004; Piotrowska et al., 2017). The mean delay per mix $1/\lambda_M$ is the central design parameter of the system and is varied across the experiments. The sender delay is coupled to it through the ratio $\rho = \lambda_S/\lambda_M$, i.e. $1/\lambda_S = (1/\lambda_M)/\rho$, for two reasons. Firstly, this ratio is the quantity that the security analysis of continuous mix networks constrains, as the unlinkability bound of Das et al. (2024) holds for a constant ratio of the sender rate to the mix rate rather than for a fixed sender delay. Secondly, a sender delay fixed at an absolute value would exceed the mixing contribution to $\text{Var}(Z)$ when the mix delay is small, and the low end of every curve would then measure the sender rather than the mixes. The ratio is set to $\rho = 4$. A sender delays only its own message, which widens the timing spread but mixes nothing. The sender delay must therefore remain a minor share of the total variance, or the measured anonymity would be attributed to a mechanism that cannot provide it. Within the range $\rho \geq 4$, in which this holds, $\rho = 4$ is the most conservative choice, because at a fixed total expected delay the variance of Z increases with ρ .

The total delay Y from the start of slot t to the exit of a message emitted by sender i in that slot and delivered to destination r is the sum of the path latency of equation (3.16), the intentional delay of equation (3.17), and the jitters:

$$Y = \mu(i, r) + Z + \sum_{l=1}^{L+1} \varepsilon_l. \quad (3.18)$$

The trace records one entry event and one exit event per message. The entry records the sender and its slot t , and the exit records the destination r and the exit time $t + Y$. The realised delays and jitters appear in no record. Because the exit of the real message is followed by the public broadcast of the block, the system does not conceal which destination received the genuine proposal. What it conceals is the correspondence between that exit and a sender in S_t , and how well the delays and latencies defined here conceal it is the subject of the following chapters.

3.4 THE STRATIFIED MIX NETWORK

Deployed mix networks do not route every message through one chain of mixes. They arrange the mixes in L layers of W nodes, and a message traverses one node of every layer (Piotrowska et al., 2017; Diaz et al., 2021). This stratified topology serves two purposes. It divides the load of approximately $p_s N$ messages per slot among W nodes rather than one, and it provides each message with one of several routes, which the GPA does not observe. Hence, the timing of a message depends on an unobserved route as well as on the random delays. The stratified mix network of this dissertation is the single-path mix generalised to this topology. At $W = 1$ it reduces to the single path, and every other component is unchanged, so that any difference between the results of the two AC systems is attributable to the topology.

The model, shown in Figure 3.9, comprises the mix nodes $M_{j,w}$ for $j = 1, \dots, L$ and $w = 1, \dots, W$, where j is the layer and w the position within the layer. Every node of layer j is connected to every node of layer $j + 1$. A message enters at a node of the first layer and is forwarded layer by layer. The sequence of nodes it visits is its route π , written as the positions w_j of those nodes:

$$\pi = (w_1, \dots, w_L), \quad S_t \rightarrow M_{1,w_1} \rightarrow M_{2,w_2} \rightarrow \dots \rightarrow M_{L,w_L} \rightarrow R_t. \quad (3.19)$$

From the last layer, the message leaves for its destination in R_t . Because each w_j takes one of W values, the network contains W^L distinct routes, and the single path of equation (3.13) is the case $W = 1$ with only one route.

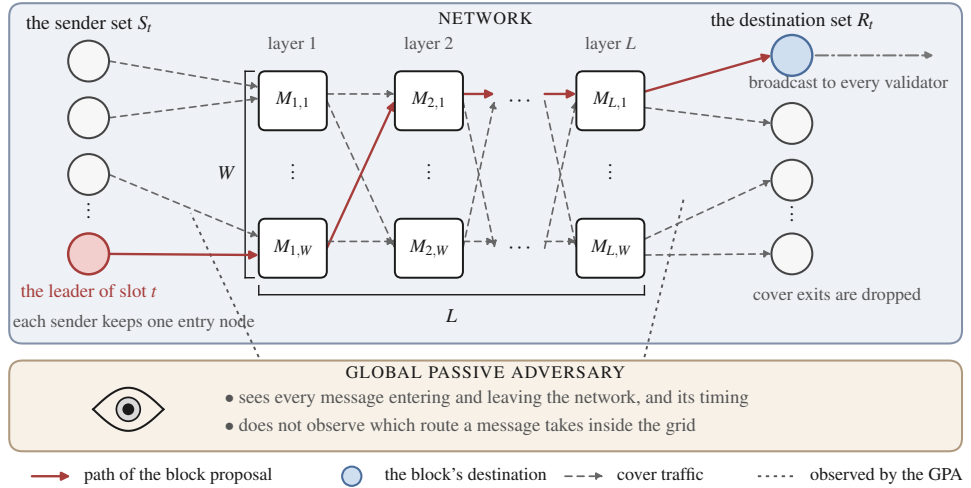


Figure 3.9: The stratified mix network. The mixes form L layers of W nodes, and every node of a layer is connected to every node of the next. Each sender is attached to one node of the first layer, and each of its messages takes a route chosen at random through the remaining layers. The route is not observed by the GPA.

The route of a message is determined in two steps. Firstly, the N validators are

divided evenly among the W nodes of the first layer, and each keeps its entry node for the whole epoch, so that w_1 is fixed for every message of a given validator. Secondly, the remaining nodes $w_2, \dots, w_L$ are chosen uniformly at random for every message. A validator's messages may thus take any of the W^{L-1} routes through its entry node. This assignment is called the split rule. Under the alternative, the uniform rule, the entry node is also chosen uniformly at random for every message, and a message may take any of the W^L routes. The split rule represents the gateway layer of deployed mix networks, in which a client is attached to one provider or gateway and only the route through the mix layers is chosen per message (Piotrowska et al., 2017; Diaz et al., 2021). The model does not simulate this layer, and the fixed attachment stands in for it. An explicit gateway layer can be added should a designer require one. The attachment is part of the topology and is therefore public, which reduces the routes that a message could have taken from W^L to W^{L-1} .

Relative to the single path, the only change to the timing of a message is that its deterministic path latency depends on the route. Every link of the network draws its quenched latency from the link law of equation (3.11), and equation (3.16) becomes:

$$\mu(i, \pi, r) = d_{i,w_1}^S + D_M(\pi) + d_{r,w_L}^R, \quad (3.20)$$

where $d_{i,w}^S$ is the latency from validator i to node $M_{1,w}$, $d_{r,w}^R$ that from node $M_{L,w}$ to validator r , and $D_M(\pi)$ the sum of the $L - 1$ inter-layer latencies along route π . The entry-link latency d_{i,w_1}^S is fixed for every message of sender i , whereas $D_M(\pi)$ and d_{r,w_L}^R vary with the route.

Everything else is unchanged. The total delay Y of equation (3.18) holds with $\mu(i, \pi, r)$ in place of $\mu(i, r)$. Because every route crosses L mixes and $L + 1$ links regardless of the width, W adds no bandwidth and no latency in expectation and only multiplies the routes and the mix nodes. In contrast, the depth L adds a link and a mixing delay to every message.

The network is modelled on Loopix, whose design Nym deploys, with two simplifications. Firstly, the mixes emit no loop cover of their own (Piotrowska et al., 2017). The adversary is passive and the mixes lie outside the validator set, and such cover would therefore never enter a sender set S_t . Its congestion cost is represented by the jitter of the link law, and its bandwidth is not counted, as the bandwidth overhead comprises the messages of the sender set only. Secondly, neither the mixing delay nor the jitter depends on load, so cover adds no latency by construction. Within these limits, the stratified mix network is the primary defence of this dissertation, evaluated against the single-path baseline.

3.5 THE ADVERSARY

The adversary is the GPA. It is global in that it observes every link of the network, including the links of the AC system, with accurate timestamps. It is passive in that it only records the traffic and never delays, drops, injects, or modifies a message. It is also external in that it has no access to the interior of any validator or mix node. The GPA observes two things, the trace, which records the entry and exit events of every message, and the public blockchain, on which every block states its slot. Therefore, for every block, the GPA knows the slot t of its proposal, the sender set S_t of that slot, the destination r that received the proposal, and its delay Y . The entry is recorded at the slot in which the message is sent rather than at the instant at which it leaves the sender. Hence, the sender delay X_S , the mixing delays, and with them the whole random delay Z are unobserved. This slot-resolution entry favours the defence, as the timestamp on the link from the sender would otherwise reveal X_S . Its effect is small, because the sender delay contributes only a minor share of the variance of Z by the choice of ρ . Everything inside the AC system is also hidden. The GPA cannot match a message leaving a mix node to the message that entered it (Chaum, 1981), and thus neither the route nor the realised delays and jitters are observed, and the outcome of the lottery is known only through the broadcast it causes.

The GPA knows the system in full, with the exception of the hidden quantities just listed (Piotrowska et al., 2017; Das et al., 2024). Specifically, it knows the protocol constants f , T , and N and the cover probability p_s . It knows the configuration of the AC system, i.e. L , W , the attachment of validators to entry nodes, and the rates λ_M and λ_S , which fix the distribution of the delays but not the delay drawn for any message. It also knows the link law, including the jitter rate λ_ϵ , and every quenched link latency. The stakes themselves are unknown to it, and its patience and computation are unlimited. This knowledge is the worst case for the defence, and the leakage measured against the GPA is thus an upper bound on what a less informed observer could extract from the same trace. Active, node-corrupting, and cryptographic adversaries lie outside the model.

One stronger variant is defined for the stratified mix network. Mixing conceals the route only while several messages are inside a mix node at once. When traffic is sparse relative to W , a message may cross a node on its own, and an observer of the links can trace its route. The delays of the model do not depend on load, and this tracing therefore cannot be represented directly. Instead, two adversaries bound the leakage. The GPA defined above does not observe the route and provides a lower bound. The route-oracle GPA is the same adversary given the true route of every message, and it provides an upper bound, because additional evidence cannot lower the expected accuracy of an exact posterior. Under the split rule, the route reveals the sender’s entry node and excludes the candidates attached to other entry nodes. At $W = 1$ the two adversaries coincide.

The GPA pursues two goals. The first is to identify the sender of an individual block

proposal. For this, the GPA forms a posterior over the candidates in S_t , i.e. it assigns to each candidate a probability of having sent the proposal given the observed exit, and it names the most probable one. This is the question of sender unlinkability, and the privacy target is a posterior close to uniform over S_t . The second goal is to recover the distribution of stake. By equation (3.15), a validator with more stake appears in the sender sets more often. Its stake can therefore be estimated from the frequency with which it appears over one or more epochs, and a comparison across validators may identify the largest stakeholder. This is the question of stake privacy.

3.6 LATENCY AND BANDWIDTH OVERHEAD

The anonymity provided by the AC system comes at a cost in latency and in bandwidth. Its mixing delays slow the propagation of every block proposal, and its cover traffic adds messages to every slot. Together with the anonymity itself, these two costs are the three axes of the anonymity trilemma (Das et al., 2018), and the hypothesis of this dissertation requires the leakage to be suppressed at a latency and bandwidth cost compatible with the constraints of the consensus protocol. Both costs are expressed as overhead ratios relative to the baseline without an AC system, in which the leader broadcasts its block directly through the gossip network at the start of its slot and no other message is sent.

Let Δ denote the propagation delay of a block proposal, i.e. the time from the start of its slot until the block has reached every validator. Without the AC system, $\Delta = D_{\text{br}}$. With the AC system, the broadcast begins only when the proposal leaves the path, after the delay Y of equation (3.18), and $\Delta = Y + D_{\text{br}}$. The latency overhead $\ell = \mathbb{E}[\Delta]/\mathbb{E}[D_{\text{br}}]$ is the ratio of the mean propagation delays with and without the AC system and states how many times slower block propagation becomes. Taking the expectation of equation (3.18) over the link latencies, the intentional delays, and the jitter yields:

$$\begin{aligned}\ell &= 1 + \frac{\mathbb{E}[Y]}{\mathbb{E}[D_{\text{br}}]} \\ &= 1 + \frac{(L+1)\mathbb{E}[d_e(m)] + 1/\lambda_S + L/\lambda_M}{\mathbb{E}[D_{\text{br}}]},\end{aligned}\tag{3.21}$$

where $\mathbb{E}[d_e(m)] = d_0 + \exp(a + b^2/2) + 1/\lambda_\epsilon$ is the mean latency of one link under equation (3.11), equal to 62.9 ms when the jitter is absent. The numerator comprises the mean latency of the $L+1$ links of the path, the mean sender delay $1/\lambda_S$, and the mean total mixing delay L/λ_M . The same expression holds for the stratified mix network, whose routes all comprise $L+1$ links and L mixing delays. Thus, the latency overhead rises with the depth, the link latency, and the intentional delays, whereas a larger network has a larger baseline broadcast latency $\mathbb{E}[D_{\text{br}}] = O(\log N)$, which may reduce the relative overhead.

The consensus protocol sets an upper limit on this overhead. The mean propagation

delay of a block must remain below the mean block interval (see Appendix A.2):

$$\mathbb{E}[\Delta] < \frac{1}{f} = 30 \text{ s.} \quad (3.22)$$

A block that has not reached the next leader when that leader produces its own block causes a fork, and under the longest-chain rule only one of the two blocks survives (Decker and Wattenhofer, 2013). The Logos protocol infers the total active stake from the observed chain growth (Logos, 2026a), and forks lower this estimate, raise every validator's share of it, and increase block production until the surviving growth returns to f . This compensation is possible only while $\mathbb{E}[\Delta] < 1/f$, because at the threshold block production reaches one block per slot and the inferred total stake becomes zero. Hence, equation (3.22) is a hard limit on the delay that the AC system may add, and it constrains the mean delay rather than its maximum.

The bandwidth overhead β is the number of messages that the AC system processes for every genuine block proposal. In slot t the AC system processes $|S_t|$ messages, of which n_t are genuine proposals, and without the AC system only those n_t messages are sent. Because every message has the same size, the ratio of the two counts is the ratio of the bandwidth consumed with and without the AC system. Over an epoch it is measured as $\sum_{t=1}^T |S_t| / \sum_{t=1}^T n_t$, which converges to the ratio of the expectations as $T \rightarrow \infty$. Taking the expectation of equation (3.14) in the numerator yields:

$$\begin{aligned} \beta &= \frac{\mathbb{E}[|S_t|]}{\mathbb{E}[n_t]} \\ &= \frac{\mathbb{E}[n_t] + p_s (N - \mathbb{E}[n_t])}{\mathbb{E}[n_t]} \\ &= 1 + p_s \left(\frac{N}{\mathbb{E}[n_t]} - 1 \right). \end{aligned} \quad (3.23)$$

This expression separates the contribution of the genuine block proposals, the leading term, from the additional communication induced by the slot-synchronous cover traffic. As expected, the bandwidth overhead increases linearly with the cover probability p_s and decreases as the expected number of leaders per slot $\mathbb{E}[n_t]$ increases.

The two overheads depend on different parameters. Cover traffic enters only β , because the delays of the model do not depend on load, whereas the depth and the intentional delays enter only ℓ , because β counts messages rather than link transmissions. Consequently, the two costs of the trilemma can be varied independently.

CHAPTER 4

ATTACKS, MEASURES, AND FRAMEWORK

This chapter derives the two attacks of the GPA together with the measures that evaluate their success. Stake inference threatens stake privacy, and Bayesian sender attribution threatens sender unlinkability. The experimental framework in which the attacks are run is then described.

4.1 STAKE INFERENCE AND ITS MEASURES

The stake inference attack recovers the relative stake of every validator from the frequency of its appearances in the sender sets. By equation (3.15), a validator belongs to S_t with probability q_i , which increases with its stake, and the GPA observes the sender set of every slot.

For every validator i , the GPA counts the slots of the epoch in which it appears in the sender set and forms its participation frequency, with $\mathbf{1}[\cdot]$ the indicator function:

$$c_i = \sum_{t=1}^T \mathbf{1}[i \in S_t], \quad \hat{q}_i = \frac{c_i}{T}. \quad (4.1)$$

The frequency $\hat{q}_i$ is an unbiased estimator of the participation probability q_i , i.e. $\mathbb{E}[\hat{q}_i] = q_i$. Solving equation (3.15) for $\phi(\alpha_i)$ yields the unbiased estimate of the win probability:

$$\hat{\phi}_i = \frac{\hat{q}_i - p_s}{1 - p_s}, \quad (4.2)$$

and inverting the lottery function of equation (3.1) provides the estimate of the relative stake:

$$\hat{\alpha}_i = \frac{\ln((1 - \hat{q}_i)/(1 - p_s))}{\ln(1 - f)}, \quad (4.3)$$

where f and p_s are public knowledge. Because $\hat{q}_i \rightarrow q_i$ as $T \rightarrow \infty$ by the strong law of large numbers, the estimate converges to the true stake, $\hat{\alpha}_i \rightarrow \alpha_i$. A validator whose frequency falls below the cover baseline, $\hat{q}_i < p_s$, would receive a negative estimate. This

is sampling noise rather than negative stake, and such estimates are set to zero. The two limits of the estimator follow from the form of q_i . As $p_s \rightarrow 0$, $\hat{q}_i \approx \phi(\alpha_i)$ and $\hat{\alpha}_i \approx \alpha_i$, and the stake is revealed. As $p_s \rightarrow 1$, $\hat{q}_i \approx p_s$ for every validator, the stake-dependent part $(1 - p_s) \phi(\alpha_i)$ of q_i tends to zero, and the estimate contains no information about stake.

The estimate $\hat{\alpha} = (\hat{\alpha}_1, \dots, \hat{\alpha}_N)$ is evaluated against the true stake vector by two measures. The first asks whether the GPA names the largest stakeholder:

$$P_{\max} = \mathbb{P}\left(\arg \max_i \hat{\alpha}_i = \arg \max_i \alpha_i\right). \quad (4.4)$$

One epoch yields a single success or failure, and the measure is the average over independent repetitions, against which a random guess succeeds with probability $1/N$. Because $\ln(1 - f) < 0$, the estimate of equation (4.3) increases with the count c_i , and the validator with the largest estimate is the one that sent in the most slots. Hence, $P_{\max}$ measures whether the largest stakeholder can be identified only from the sending frequencies, and the GPA need not know f or p_s to name it. Validators with the same count tie, and a shared largest estimate is scored as the probability that a uniform guess among the tied validators is correct, i.e. $1/3$ if three validators tie and the true largest stakeholder is among them, and 0 otherwise. Ties are not rare when the largest stakes are close, as the counts are integers. The measure $P_{\max}$ is the operational statement of stake privacy, because the value of p_s at which it reaches $1/N$ is the cover probability at which the largest stakeholder is hidden, and equation (3.23) states its cost.

The second measure asks the same question about the group of largest stakeholders rather than the single largest. The validators are ranked by true stake and by estimated stake, and the top fraction x of each ranking forms the true set Top_x and the estimated set $\widehat{\text{Top}}_x$, of the same size. The Jaccard index (Jaccard, 1912) compares them:

$$J_x = \frac{|\widehat{\text{Top}}_x \cap \text{Top}_x|}{|\widehat{\text{Top}}_x \cup \text{Top}_x|}, \quad (4.5)$$

i.e. the number of validators correctly placed in the group divided by the number in at least one of the two sets. Hence, $J_x = 1$ when the group is identified exactly and $J_x = 0$ when no member is identified. For example, if the GPA identifies five of the ten largest stakeholders, the two sets share five validators out of fifteen in total, and $J_x = 1/3$. In this project, the group is the top 1% of validators by stake. Unlike $P_{\max}$, which succeeds or fails in one epoch, J_x degrades gradually and therefore shows whether the GPA learns who the large stakeholders are as a group even when the single largest is not named.

4.2 BAYESIAN SENDER ATTRIBUTION AND ITS MEASURES

The Bayesian sender attribution attack identifies the sender of an individual block proposal from its observed slot t , sender set S_t , destination r , and exit time y . The exit time is the observed value of the delay Y of equation (3.18), measured from the start of the slot to the moment at which the destination initiates the broadcast. The hidden variable is the true sender $i^* \in S_t$, and the GPA computes the posterior probability that each candidate $i \in S_t$ sent the proposal.

The attack uses the timing model of the AC system. For every candidate i , the GPA subtracts the deterministic path latency $\mu(i, r)$ of equation (3.16) from the observed delay y . The residual of candidate i is the part of y that the deterministic latency does not explain:

$$z_i = y - \mu(i, r). \quad (4.6)$$

By equation (3.18), the residual of the true sender equals the residual delay $\tilde{Z} = Z + \sum_{l=1}^{L+1} \varepsilon_l$, the intentional delay plus the jitter of the $L+1$ links. Because both are non-negative, a candidate with $z_i < 0$ is impossible and is assigned likelihood zero. Otherwise, the likelihood is the probability density of the residual delay evaluated at z_i :

$$\mathcal{L}_i = f_{\tilde{Z}}(z_i). \quad (4.7)$$

The density $f_{\tilde{Z}}$ is fully determined by the parameters L , λ_S , λ_M , and λ_ε . The intentional delay Z of equation (3.17) is the sum of an exponential variable with rate λ_S and a Gamma variable with shape L and rate λ_M , and for $\lambda_S \neq \lambda_M$ its density is (see Appendix A.3):

$$f_Z(z) = A \left[e^{-\lambda_S z} - e^{-\lambda_M z} \sum_{j=0}^{L-1} \frac{((\lambda_M - \lambda_S) z)^j}{j!} \right] \quad \text{for } z \geq 0, \quad (4.8)$$

where $A = \lambda_S \lambda_M^L / (\lambda_M - \lambda_S)^L$, with mean $\mathbb{E}[Z] = 1/\lambda_S + L/\lambda_M$ and variance $\text{Var}(Z) = 1/\lambda_S^2 + L/\lambda_M^2$. The summed jitter $\sum_{l=1}^{L+1} \varepsilon_l$ is a Gamma variable with shape $L+1$ and rate λ_ε , independent of Z . Hence, the density $f_{\tilde{Z}}$ is the convolution of f_Z with this Gamma density:

$$f_{\tilde{Z}}(z) = \int_0^z f_Z(z-u) \frac{\lambda_\varepsilon^{L+1} u^L e^{-\lambda_\varepsilon u}}{L!} du. \quad (4.9)$$

Without jitter, $f_{\tilde{Z}} = f_Z$, which is the case in every experiment unless stated otherwise.

The posterior follows from Bayes' rule. With a prior $p_0(i)$ over the candidates, i.e. the probability of candidate i before the timing is observed, the posterior probability that

candidate i sent the proposal is:

$$p(i) = \mathbb{P}(i^* = i \mid y, r, S_t) = \frac{p_0(i) \mathcal{L}_i}{\sum_{j \in S_t} p_0(j) \mathcal{L}_j}. \quad (4.10)$$

In the absence of further information, the prior is uniform, $p_0(i) = 1/|S_t|$, and cancels, leaving $p(i) = \mathcal{L}_i / \sum_{j \in S_t} \mathcal{L}_j$. The GPA names the most probable candidate, the maximum a posteriori (MAP) estimate:

$$\hat{i} = \arg \max_{i \in S_t} p(i), \quad (4.11)$$

and the posterior probability $p(\hat{i})$ is its confidence in that choice. Because the mix chain is shared and the destination is observed, the terms D_M and d_r^R of equation (3.16) are common to every candidate. Thus, the residual of candidate i is $z_i = \tilde{Z} + (d_{i^*}^S - d_i^S)$, and only the differences between the sender-link latencies d_i^S separate the candidates. If all candidates have the same d_i^S , every residual is equal, and the posterior is uniform. The GPA applies these steps to every observed block proposal.

In the stratified mix network, the GPA does not observe the route π of a message, on which the deterministic latency $\mu(i, \pi, r)$ of equation (3.20) depends. The GPA therefore averages the likelihood over the routes that the message could have taken. Under the split rule, a message of validator i takes each of the W^{L-1} routes through its entry node with equal probability, and the likelihood becomes:

$$\mathcal{L}_i = \frac{1}{W^{L-1}} \sum_{\pi} f_{\tilde{Z}}(y - \mu(i, \pi, r)), \quad (4.12)$$

where the sum runs over the routes through the entry node of i . The posterior and the MAP estimate follow from equations (4.10) and (4.11) unchanged, and at $W = 1$ equation (4.12) reduces to equation (4.7). For the route-oracle GPA, which knows the true route π^* , the likelihood is the single term $f_{\tilde{Z}}(y - \mu(i, \pi^*, r))$ for the candidates attached to the entry node of π^* , and zero for every other candidate.

The attack is run on every genuine block proposal of the epoch, $b = 1, \dots, B$. For proposal b , let i_b^* be the true sender, $\hat{i}_b$ the MAP estimate, and $p_b(\cdot)$ the posterior over its sender set S_{t_b} . Three measures are computed by averaging over the proposals. The first is the top-1 de-anonymisation probability, the fraction of proposals whose sender the GPA names correctly:

$$P_{\text{Top1}} = \frac{1}{B} \sum_{b=1}^B \mathbf{1}[\hat{i}_b = i_b^*]. \quad (4.13)$$

The second, the mean true posterior, is the average probability that the GPA assigns to the true sender:

$$\bar{P}_{\text{true}} = \frac{1}{B} \sum_{b=1}^B p_b(i_b^*). \quad (4.14)$$

The third, the expected posterior entropy, is the average remaining uncertainty of the GPA about the sender, in bits:

$$\bar{H} = \frac{1}{B} \sum_{b=1}^B H_b, \quad H_b = - \sum_{i \in S_{t_b}} p_b(i) \log_2 p_b(i). \quad (4.15)$$

For every proposal, $0 \leq H_b \leq \log_2 |S_{t_b}|$. The upper bound is attained by the uniform posterior, $p_b(i) = 1/|S_{t_b}|$, under which the GPA has learnt nothing from the timing, and the lower bound when the posterior concentrates on a single candidate. Under the uniform posterior, the top-1 probability and the true posterior both equal $1/|S_{t_b}|$, the probability of a random guess. Because $|S_t|$ varies from slot to slot, these reference values are averaged over the proposals of the epoch in the same way as the measures, i.e. the random-guess probability is $\mathbb{E}[1/|S_t|]$ and the entropy ceiling is $\mathbb{E}[\log_2 |S_t|]$ under the law of equation (3.14). The top-1 probability is the operational statement of sender unlinkability, whereas the mean true posterior and the entropy describe the shape of the whole posterior and do not depend on which candidate ranks first.

The effectiveness of the attack is governed by the competition between the deterministic latency differences of the candidates, the signal, and the random delays, the noise. For a slot with sender set S_t , the mean and the standard deviation of the latencies d_i^S of the candidates are:

$$\bar{d} = \frac{1}{|S_t|} \sum_{i \in S_t} d_i^S, \quad \sigma_d = \sqrt{\frac{1}{|S_t|} \sum_{i \in S_t} (d_i^S - \bar{d})^2}. \quad (4.16)$$

If the candidates have nearly identical deterministic latencies, $\sigma_d \approx 0$, the timing provides little information about the sender. The noise is the residual delay $\tilde{Z}$, whose terms are independent, so that its variance is the sum of the variances of the intentional delay and of the summed jitter:

$$\sigma_{\tilde{Z}}^2 = \sigma_Z^2 + \sigma_\varepsilon^2, \quad \text{where} \quad \sigma_Z^2 = \frac{1}{\lambda_S^2} + \frac{L}{\lambda_M^2} \quad \text{and} \quad \sigma_\varepsilon^2 = \frac{L+1}{\lambda_\varepsilon^2}. \quad (4.17)$$

The dimensionless latency parameter is the ratio of the signal to the noise:

$$\eta = \frac{\sigma_d}{\sigma_{\tilde{Z}}} = \frac{\sigma_d}{\sqrt{1/\lambda_S^2 + L/\lambda_M^2 + (L+1)/\lambda_\varepsilon^2}}. \quad (4.18)$$

Without jitter, $\eta = \sigma_d/\sigma_Z$. When $\eta \ll 1$, the random delays exceed the deterministic latency differences, the posterior remains close to the prior, and the GPA performs only slightly better than random guessing. When $\eta \approx 1$, the two are of comparable magnitude, and the timing begins to reveal the sender. When $\eta \gg 1$, the latency differences exceed the random delays, and the posterior concentrates on the true sender. Hence, P_{Top1} and $\bar{P}_{\text{true}}$ are expected to increase with η , whereas $\bar{H}$ is expected to decrease. In the stratified mix network, d_i^S is replaced by the latency of the candidate's fixed entry link, d_{i,w_1}^S . Because S_t

changes from slot to slot, σ_d is computed per sender set and averaged over the proposals of the epoch, and η is evaluated for every configuration.

4.3 THE EXPERIMENTAL FRAMEWORK

The attacks and measures are evaluated within the simulation framework of Figure 3.1, which is modular. Each stage of the pipeline is a component with a fixed interface, and any of them can be replaced by another of the same kind without changing the rest. Consequently, both AC systems are evaluated by the same attack and the same measures, and a new AC system, attack, or measure is added as one further component. This modularity makes the framework an analytical and computational tool for the designer of an AC system. In addition, the experiments are seeded for reproducibility. The framework is implemented in Python with NumPy (Harris et al., 2020) and SciPy (Virtanen et al., 2020).

The constants of the baseline configuration are already fixed, and the one remaining parameter is the size of the network. The Logos Blockchain has no operating network yet and provides no validator count to adopt. Therefore, the reference is Cardano, with roughly 2,700 stake pools (Koios, 2026). The size $N = 1000$ is of the same order as this reference and keeps the computational cost of a full-epoch simulation moderate.

CHAPTER 5

RESULTS

The two attacks read different parts of the trace, i.e. which validators send and when their messages leave the AC system. Therefore, the results are reported by attack, each with its discussion, its cost, and the parameters it fixes.

5.1 STAKE INFERENCE

Stake inference is evaluated by $P_{\max}$ and J_x , applied to the estimate $\hat{\alpha}$ formed over one epoch. The attack reads only the sequence of sender sets S_t , which the AC system does not conceal. Hence, the results are independent of the AC system and hold identically for the single-path mix and the stratified mix network, and cover traffic is the only defence against the attack.

Both measures have an exact closed form, against which the simulation is checked. Because the largest estimate is the largest count, $P_{\max}$ is determined by the order statistics of the counts, which are independent Binomial(T, q_i) variables. Let $i_{\max} = \arg \max_i \alpha_i$ be the largest stakeholder. Conditioning on its count c , the GPA names it if every other count is at most c , and an epoch in which n other validators tie at c scores $1/(n+1)$ under the tie rule of equation (4.4). Writing this factor as $\int_0^1 u^n du = 1/(n+1)$ yields:

$$P_{\max}(\alpha) = \sum_{c=0}^T \mathbb{P}(c_{i_{\max}} = c) \int_0^1 \prod_{i \neq i_{\max}} [\mathbb{P}(c_i < c) + u \mathbb{P}(c_i = c)] du, \quad (5.1)$$

where the integral in u is a polynomial of degree equal to the number of validators that can tie. The expression is evaluated numerically by Gaussian quadrature in u over the counts c at which the summand is not negligible, and the reported value is its average over stake vectors drawn from the Pareto law.

The second measure asks the same question about the group of the ten largest stakeholders, the top 1% at $N = 1000$. With $I = |\widehat{\text{Top}}_x \cap \text{Top}_x|$ the overlap of the two sets, the union has $2m - I$ members for a set size m , and $J_x = I/(2m - I)$. Because J_x is convex

in I , its expectation requires the distribution of the overlap rather than its mean:

$$\mathbb{E}[J_x \mid \boldsymbol{\alpha}] = \sum_{\kappa=0}^m \mathbb{P}(I = \kappa) \frac{\kappa}{2m - \kappa}. \quad (5.2)$$

The law of I follows by conditioning on the m -th largest count, the threshold. Given a threshold c , the estimated set comprises every validator whose count exceeds c and a uniformly random subset of those whose count equals c , as in the tie term of equation (5.1). As the counts are independent, the joint law of the numbers of validators above and at the threshold is computed separately for the true top group and for the remaining validators. The two laws are then combined over every threshold.

Figure 5.1 shows both measures against the cover probability for three Pareto shapes. Two trends are visible. Firstly, both measures decrease as p_s increases, for every shape,

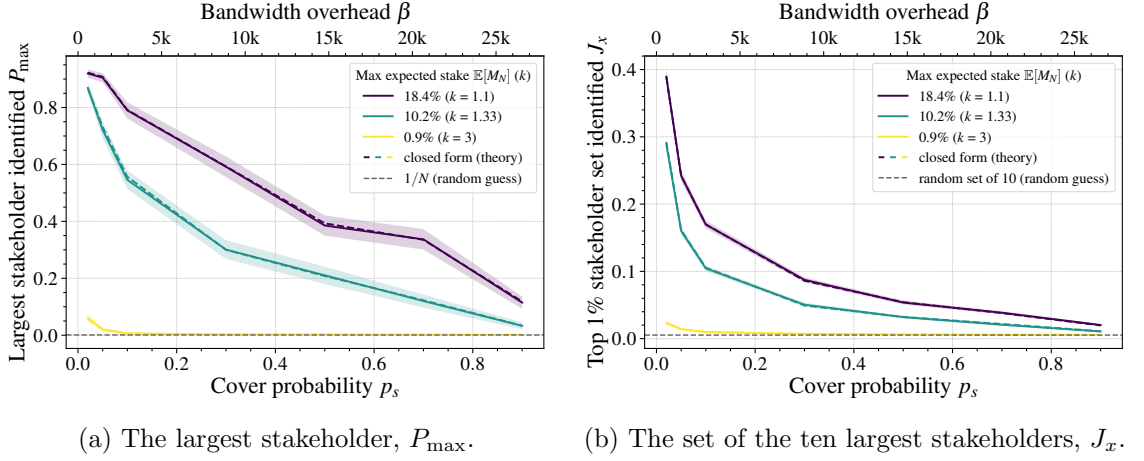


Figure 5.1: (a) The probability that the largest stakeholder is identified and (b) the Jaccard index between the estimated and the true set of the ten largest stakeholders, against the cover probability (bottom axes) and the corresponding bandwidth overhead (top axes), for three Pareto shapes labelled by their expected maximum relative stake. Solid lines are the simulation mean over 120 stake vectors of 25 epochs each, with a band of one standard error of the mean (SEM). Dashed lines are the closed forms evaluated on the same stake vectors. Horizontal dashed lines are the random-guess probability $1/N$ in (a) and the value of a uniformly random set of ten validators in (b).

because the stake-dependent term of the participation probability of equation (3.15) decreases while the error of the estimate increases. However, the decrease is slow. At the Cardano-calibrated shape, neither measure reaches its random-guess baseline, and even at high p_s the largest stakeholder is still identified. As each increment in p_s adds messages to every slot, this gradual reduction is costly. Secondly, the curves are ordered by shape. A larger k concentrates less stake in the largest stakeholder, i.e. a smaller expected maximum relative stake, and the count of the largest stakeholder therefore differs less from the others. Hence, the smaller the expected maximum relative stake, the lower the curve starts. At $k = 3$ both measures approach their baselines at moderate cover probabilities.

The group measure declines more gradually than $P_{\max}$, perhaps because it accumulates a small overlap over ten validators where $P_{\max}$ succeeds or fails on one. Throughout, the closed forms agree with the simulation to within its uncertainty.

The results at a concentrated stake are a property of the stake distribution rather than a weakness of the defence. The attack succeeds when the count of the largest stakeholder exceeds the others by a margin, and the shape affects this margin mainly through the share of the largest stakeholder. This share is the maximum relative stake $M_N = \max_i \alpha_i$, and its expectation $\mathbb{E}[M_N]$ is the quantity by which the curves are ordered. For $k > 1$ and large N , extreme-value theory yields (Arnold, 2015):

$$\mathbb{E}[M_N] \approx \frac{k-1}{k} N^{1/k-1} \Gamma\left(1 - \frac{1}{k}\right), \quad (5.3)$$

which tends to $1/N$ as $k \rightarrow \infty$ and to 1 as $k \rightarrow 1$ (see Appendix A.4). Figure 5.2 compares this approximation with a finite-sample simulation at $N = 1000$. The two agree for light tails, but the approximation overstates the value for heavy ones, because the largest stake also appears in the denominator of M_N . Hence, the finite-sample values are the ones that label the shapes. At the Cardano-calibrated shape, one validator in a thousand holds a tenth of all stake on average, and this concentration is what separates the curves.

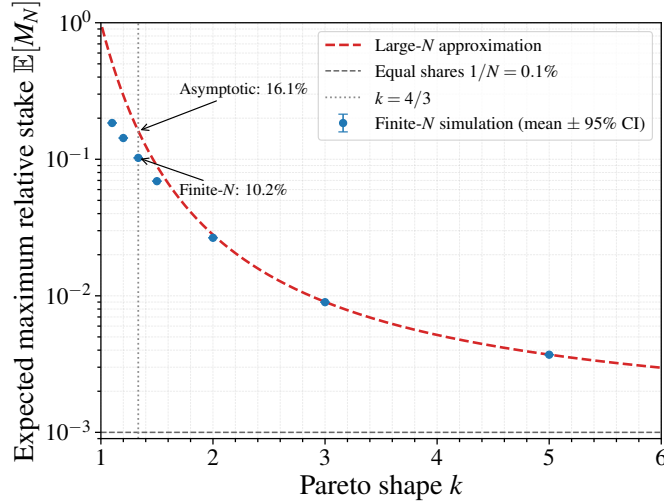


Figure 5.2: The expected share of the largest stakeholder against the Pareto shape at $N = 1000$. The dashed curve is the large- N approximation of equation (5.3), the markers are the finite-sample simulation (mean over 2,000,000 stake vectors, whose 95% confidence interval is smaller than the marker), and the horizontal line is the equal share $1/N$.

How large a share the counts can resolve follows from the precision of the estimate. The count c_i is a Binomial(T, q_i) variable with $\text{Var}(\hat{q}_i) = q_i(1 - q_i)/T$, and the delta method applied to equation (4.3) yields the standard error of the stake estimate:

$$\text{SE}(\hat{\alpha}_i) \approx \frac{1}{\lambda} \sqrt{\frac{q_i}{T(1 - q_i)}}. \quad (5.4)$$

Under cover, $q_i \approx p_s$ for every validator, and the standard error is nearly the same for all of them, about 0.016 at a representative $p_s = 0.1$. The expected largest share at $k = 4/3$, 10.2% in Figure 5.2, lies six standard errors above this noise, whereas the 0.9% at $k = 3$ lies within it, which is perhaps why the curves at $k \leq 4/3$ remain far above the baseline while the curve at $k = 3$ approaches it. This suggests that cover becomes effective once $\mathbb{E}[M_N]$ decreases to about 1% at $N = 1000$, which corresponds to $k = 3$ by Figure 5.2.

The same distribution explains the spread across stake vectors. In the finite-sample simulation of Figure 5.2, the median of M_N at $k = 4/3$ is 6.5% against a mean of 10.2%, with a standard deviation of 11 percentage points across the stake vectors. The skew arises because occasional draws contain one validator holding a large fraction of all stake, which is identified in nearly every epoch, whereas the largest stakeholder of a typical draw is not. Therefore, the outcome for a given chain depends mainly on its own stake vector.

The shape is a property of the chain rather than of the AC system, and only the protocol can change it. By the product form of the lottery of equation (3.1), the probability that a stakeholder wins a slot depends only on its total stake. A stakeholder that divides its stake among several validators therefore keeps the same probability of winning, while each of its validators holds a smaller share, which reduces the maximum relative stake. This suggests that a limit on the stake per validator is the protocol parameter that sets the Pareto shape of a chain.

5.1.1 RECOMMENDATION

Two parameters determine how much stake inference reveals. The shape of the stake distribution is a requirement on the chain, whereas the cover probability p_s is a parameter of the AC system itself that costs bandwidth.

From the point of view of the designer of an AC system, the requirement on the chain comes first, and its value involves a trade-off. A lower limit reduces the bandwidth that the AC system requires for cover traffic, but it increases the number of validators that a large stakeholder has to operate. A limit is represented here by the Pareto shape at which $\mathbb{E}[M_N]$ equals the limit, rather than by a capped distribution. At a limit of 1%, i.e. $k = 3$ at $N = 1000$, both measures approach their baselines at $p_s \approx 0.3$, and the largest stakeholder of the calibrated model, with a tenth of all stake, must operate ten validators. The baselines are the values of a random guess and cannot be lowered further. Once they are reached, a lower limit, i.e. a higher k , only reduces the bandwidth of the AC system, whereas the number of validators keeps increasing. Therefore, a limit of about 1% of the total stake per validator, $k \geq 3$ at $N = 1000$, is recommended.

The value of p_s is then set at the recommended shape, and it is a trade-off between privacy and bandwidth. Because the baseline is approached only gradually at $k = 3$, p_s cannot be chosen as the value at which identification stops. It has to be chosen as the value beyond which a further unit of bandwidth yields too little. Figure 5.3 shows both

measures over a finer grid of values of p_s . The return has no threshold at which it drops abruptly. Instead, it diminishes smoothly, and beyond $p_s = 0.1$ each further step yields little. Therefore, $p_s = 0.1$ is recommended, at a cost of $\beta \approx 2,950$ messages per proposal, or 38,880 messages per validator per epoch.

Whether this cost is affordable depends on the bandwidth available to the AC system, which is a decision for its designer. Hence, the recommended value is the most that is worth spending, and a designer with less bandwidth may choose a lower one. Figure 5.3 informs this decision, as the designer can quantify the trade-off from it, i.e. the anonymity obtained for a given bandwidth, or the bandwidth required for a given anonymity.

The recommendation guarantees a bounded residual rather than an absence of identification. At $k = 3$ and $p_s = 0.1$ the largest stakeholder is identified in roughly 1% of epochs, against about 60% at the Cardano-calibrated shape, which is why the limit on the stake is the first requirement. Moreover, the residual is a probability per epoch, and a patient adversary observes many.

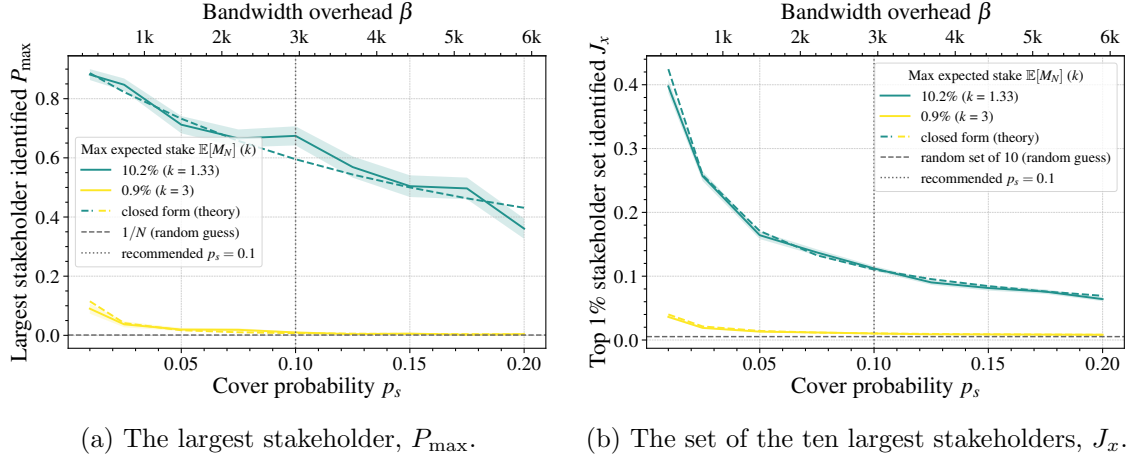


Figure 5.3: Both measures against the cover probability at $k = 4/3$ and $k = 3$. Solid lines are the simulation mean over 120 stake vectors of 25 epochs each, with a band of one SEM. Dashed lines are the closed forms of equations (5.1) and (5.2) averaged over 400 independent stake vectors. Horizontal dashed lines are the random-guess baselines, and the vertical dotted line marks the recommended $p_s = 0.1$.

5.1.2 EXTENSION TO MANY EPOCHS

The analysis is now extended from one epoch to many. For a fixed stake vector, successive epochs are independent repetitions of the same experiment. Hence, the probability of identification accumulates under continued observation. This accumulation is evaluated in three ways. Firstly, the simulation provides it directly for up to 25 epochs, the number observed for each of the 120 stake vectors. Secondly, because equation (5.1) yields $P_{\max}(\alpha)$ exactly for any stake vector, compounding it extends the accumulation to any horizon. The GPA is assumed to act on the guess of every epoch, and the quantity of interest is thus its

first correct guess. The probability that the largest stakeholder has been identified at least once within E epochs, averaged over stake vectors drawn from the Pareto distribution, is:

$$C(E) = \mathbb{E}_{\alpha} \left[1 - (1 - P_{\max}(\alpha))^E \right], \quad (5.5)$$

which is evaluated on 1,000 stake vectors. This is the quenched form, as each stake vector is fixed and compounded on its own. Thirdly, the annealed form compounds the mean rate instead:

$$1 - (1 - \mathbb{E}_{\alpha}[P_{\max}(\alpha)])^E, \quad (5.6)$$

which corresponds to a stake vector that is redrawn every epoch. Because $1 - (1 - p)^E$ is concave in p , Jensen's inequality places the annealed form of equation (5.6) above $C(E)$ at every $E > 1$. It overstates the adversary, as it assigns to every chain the per-epoch probability of the few chains whose largest stakeholder is exposed, and it is reported as an upper bound only. The quenched form is the relevant one, and the simulation checks it.

Figure 5.4 shows the three at the recommended $p_s = 0.1$ for $k = 3$ and for the Cardano-calibrated $k = 4/3$, over one year and over ten years of observation. The simulation agrees with the quenched form over its 25 epochs, and the annealed form lies above both. At $k = 4/3$ the largest stakeholder is identified with probability 0.6 in the first epoch, 0.9 within about six weeks, and almost certainly within a year, whereas at $k = 3$ the probability is about 0.3 after one year and 0.9 after about eight. Hence, the recommended shape postpones the identification rather than preventing it.

This accumulation has two limitations. Firstly, the stake vector is assumed to be fixed, whereas the stake of a real chain changes through delegation and rewards, and the largest stakeholder of one epoch is not necessarily that of a later one. The accumulation is therefore bounded by the time for which the stake distribution persists, and for a chain whose stake changes within years the curves overstate the adversary at such horizons. Secondly, Bayesian sender attribution does not compound in this way, because the evidence about the sender of a proposal is complete when it is sent. The aggregation of evidence across proposals is the long-term disclosure attack, which lies outside this dissertation.

5.2 BAYESIAN SENDER ATTRIBUTION

Bayesian sender attribution is evaluated by P_{Top1} , $\overline{P}_{\text{true}}$, and $\overline{H}$, applied to every block proposal of one epoch. The attack reads the time at which a proposal leaves the AC path, together with the sender set of its slot. Hence, the parameters of the AC system that shape this time are the defence against it.

The mean delay per mix $1/\lambda_M$ is examined from 10^{-4} s to 1 s on the single-path mix at depths $L = 1$ to 4, with cover fixed at 19 senders per slot rather than drawn with p_s , so that the candidate set has a near-deterministic size. Figure 5.5 shows the measures

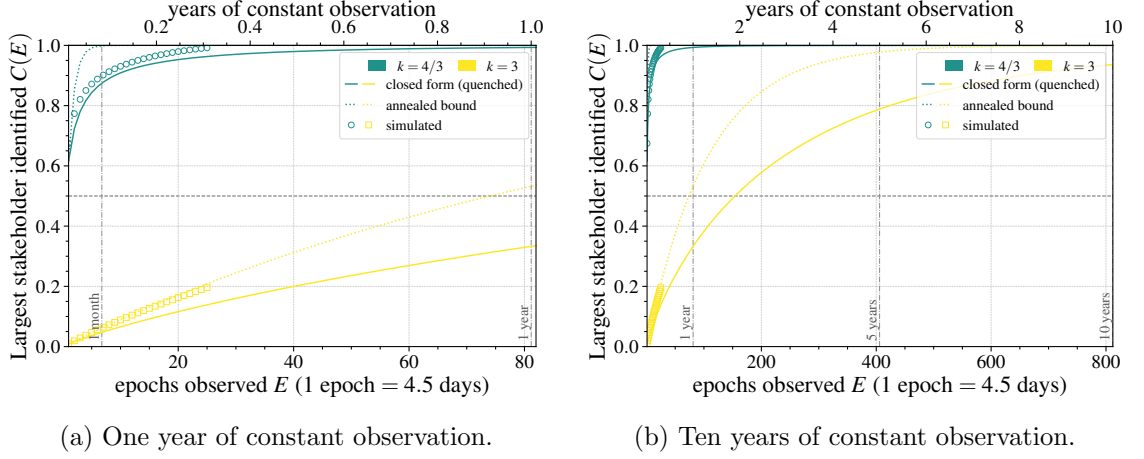


Figure 5.4: The probability that the largest stakeholder has been identified within E epochs over (a) one year and (b) ten years of constant observation at a cover probability of 0.1. Solid lines are the closed form of equation (5.5) averaged over 1,000 stake vectors, dotted lines are the annealed bound, and the markers are the simulated estimate for $E \leq 25$ from the 120 stake vectors of 25 epochs each. The dashed horizontal line marks a probability of 0.5.

against $1/\lambda_M$, one curve per depth. As expected, the identification probability decreases and the posterior entropy increases as $1/\lambda_M$ increases. A larger random delay makes it more difficult for the adversary to relate the observed exit time to the latency of any one candidate. Without intentional delay the sender is identified with certainty, and from $1/\lambda_M \approx 0.1$ s the identification probability plateaus close to the random guess. The delay influences the anonymity through η . By equation (4.17), σ_Z is proportional to $1/\lambda_M$, whereas the signal σ_d is fixed by the link law. Hence, $1/\lambda_M$ is the parameter that the designer sets, but the depth and the jitter of the network change η as well.

The depth has the same effect. At a given $1/\lambda_M$, a deeper path is more anonymous, because every layer adds a mixing delay to the noise and σ_Z increases with L . At matched η the four curves nearly coincide, which suggests that on the single path, where there is one route regardless of L , the effect of the depth arises from the noise only. The difference that depth makes is therefore quantitative rather than qualitative, as the same anonymity is obtained from a longer delay at fewer mixes. However, each layer also adds one link and one mean delay to the AC path, as Figure 5.5d shows. The value of depth may lie instead in the tolerance to compromised mix nodes, which the model does not represent.

The stratified mix network adds two parameters, the width W and the entry rule. The effect of the entry rule depends on W , and the two are therefore examined together. Figure 5.6 compares the two rules. The uniform rule is the more anonymous of the two up to $1/\lambda_M \approx 0.1$ s, beyond which the rules coincide close to the random guess. Two mechanisms separate them. Firstly, the sender link of a candidate under the split rule is one quenched link, whereas under the uniform rule it is the average of the W links to the entry nodes, and its spread, the signal σ_d , is smaller by a factor of $\sqrt{W}$. Secondly,

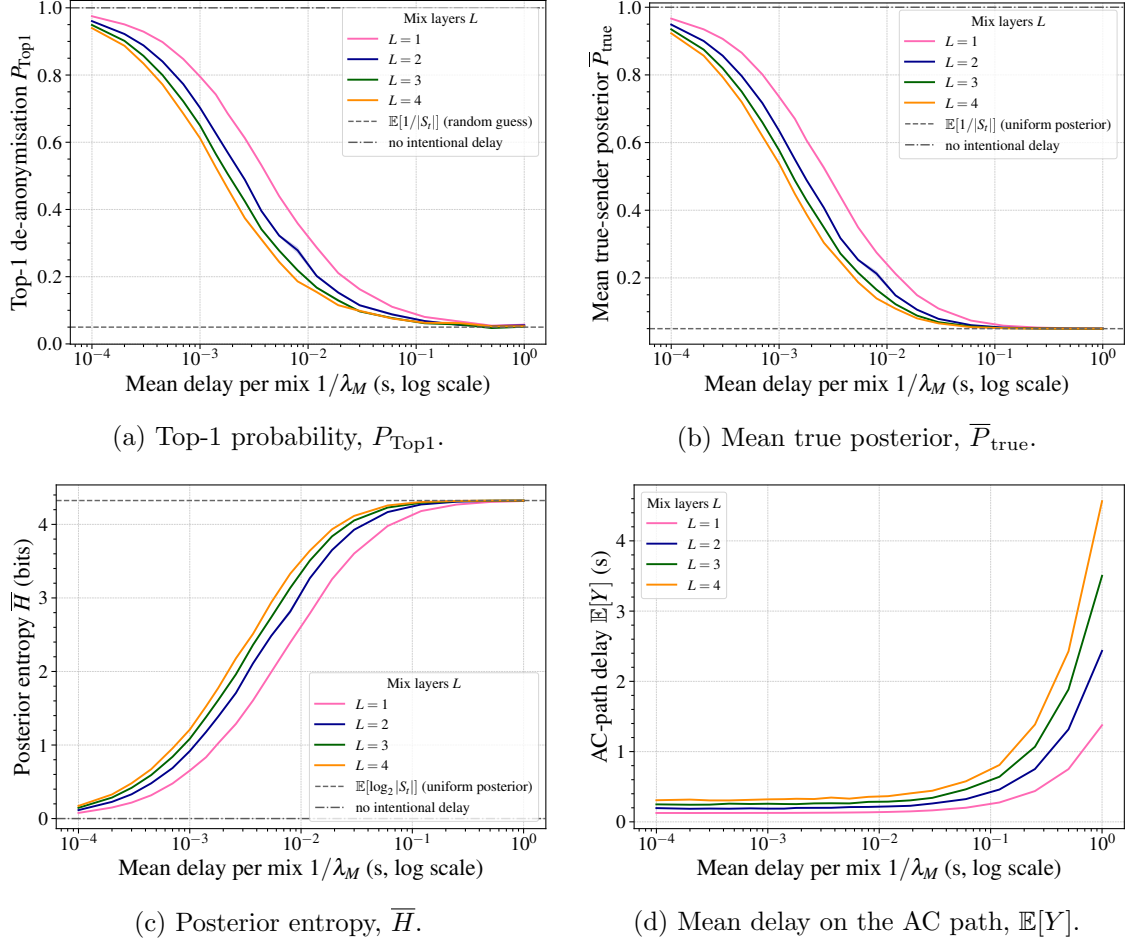


Figure 5.5: (a) The probability that the sender is named, (b) the mean posterior probability of the true sender, (c) the posterior entropy, and (d) the mean delay on the AC path from the start of the slot to the exit of the proposal, against the mean delay per mix for the single-path mix, one curve per depth L , with cover fixed at 19 senders per slot. Lines are the simulation mean over 50 epochs. Dashed horizontal lines are the values of the uniform posterior and dash-dotted lines are the limit of no intentional delay.

the entry node is public under the split rule, and the likelihood of equation (4.12) is thus averaged over W^{L-1} routes rather than W^L , which makes the posterior sharper. Both mechanisms increase with W . Figure 5.7 therefore compares the two rules against the width, showing σ_d and P_{Top1} at a fixed $1/\lambda_M = 10^{-4}$ s. Figure 5.7a validates the first mechanism. Under the split rule, σ_d remains at the spread of one link at every width, whereas under the uniform rule it decreases in proportion to $1/\sqrt{W}$. In Figure 5.7b the identification probability decreases with W under both rules.

The split rule is retained for the remaining results. Besides being the model of the gateway layer, it is the rule that favours the adversary through both mechanisms. Hence, every result reported under it is a lower bound on the anonymity of the uniform rule, and a designer who adopts the uniform rule gains the difference shown in Figure 5.7.

The width is examined next. Figure 5.8 shows P_{Top1} against $1/\lambda_M$ for several widths,

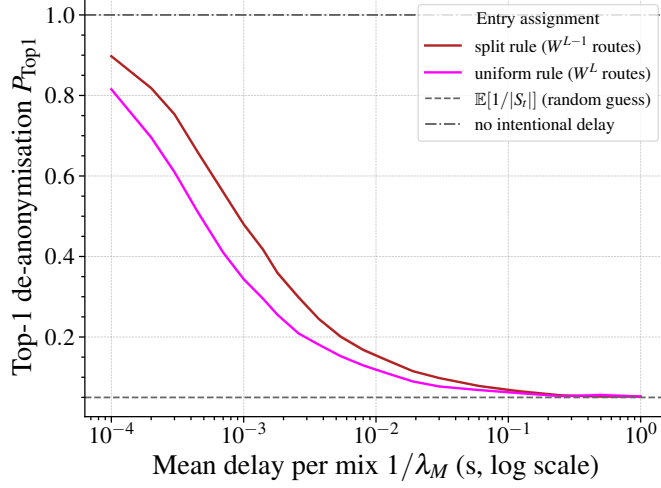


Figure 5.6: The probability that the sender is named against the mean delay per mix for the stratified mix network at $W = 2$ and $L = 3$, under the split and the uniform rules, with cover fixed at 19 senders per slot. Lines are the simulation mean over 50 epochs. The dashed horizontal line is the random-guess probability and the dash-dotted line the limit of no intentional delay.

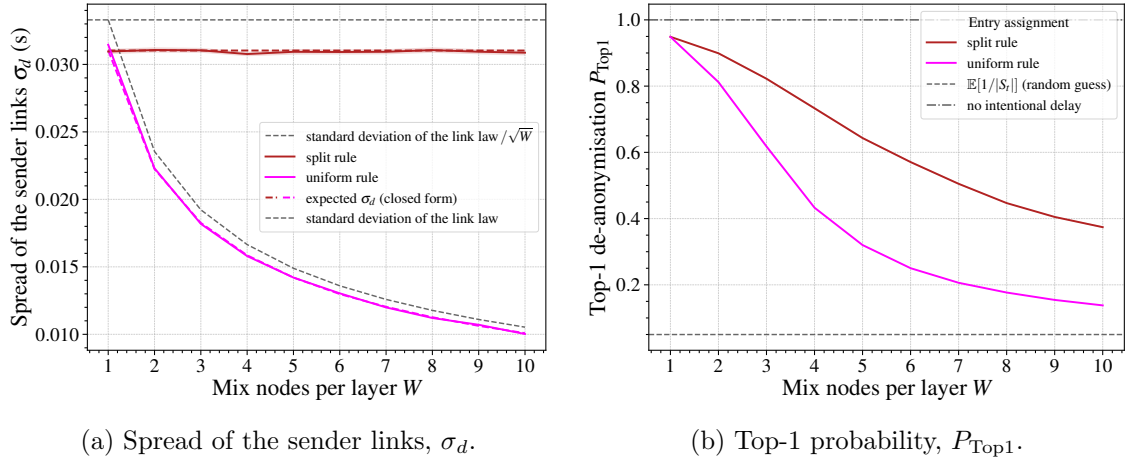


Figure 5.7: (a) The spread of the sender links and (b) the probability that the sender is named against the width W under the split rule (W^{L-1} routes per message) and the uniform rule (W^L routes), at $L = 3$, a mean delay per mix of 10^{-4} s, and cover fixed at 19 senders per slot. Lines are the simulation mean over 50 epochs. In (a) the grey dashed lines are the standard deviation of the link law, 0.033 s, and the same divided by $\sqrt{W}$. The coloured dashed lines are the expected σ_d over a sender set of about 20 candidates. In (b) the dashed line is the random-guess probability and the dash-dotted line the limit of no intentional delay.

where $W = 1$ is the single path and $W = 40$ the width of the deployed Nym network. Width reduces the identification probability where the delay is small, and the curves converge as the delay increases. This reduction costs no latency, because every route crosses $L + 1$ links and L mixes regardless of W .

The effect of width arises from the routes. In a wider network each candidate has more

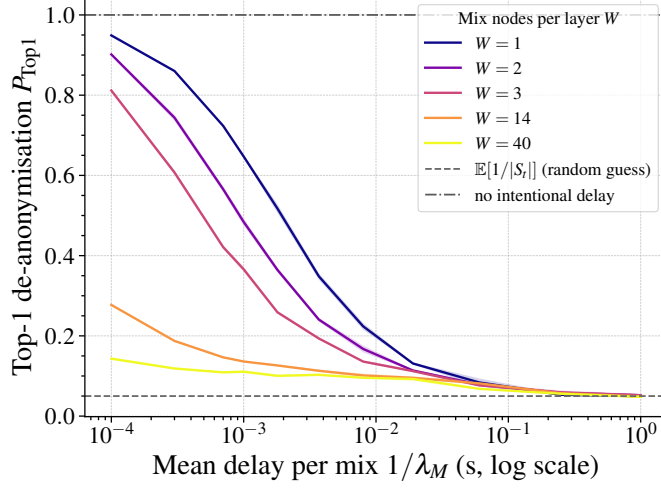


Figure 5.8: The probability that the sender is named against the mean delay per mix for the stratified mix network at $L = 3$ under the split rule, one curve per width W , with cover fixed at 19 senders per slot. Lines are the simulation mean over 30 epochs. The dashed horizontal line is the random-guess probability and the dash-dotted line the limit of no intentional delay.

routes, and thus more latencies that could have produced the observed time. Where the delay is too small to conceal these latencies, the additional routes widen the likelihood of every candidate. Once the noise exceeds the differences between the latencies of the candidates, the routes add nothing further, and the anonymity is determined by the delay. Consequently, width is primarily a parameter of capacity and robustness rather than of anonymity. Every message crosses each layer, and the load per mix node is therefore about $p_s N/W$ messages per slot. Width also dilutes the compromise of mix nodes in the same way as depth does (Diaz et al., 2021). Hence, capacity and tolerance to compromised nodes both require W to be greater than one, at the cost of infrastructure rather than latency.

The link jitter is the noise of the network itself. Every link delays a message by an additional random time with mean $1/\lambda_\epsilon$, which is a property of the network rather than a choice of the designer. It is examined on the single path at three mean delays per mix, and Figure 5.9 shows P_{Top1} and the mean delay on the AC path against $1/\lambda_\epsilon$. Without mixing, the jitter is the only noise, and it reduces the identification probability from certainty to the random guess by itself, with the plateau from $1/\lambda_\epsilon \approx 0.1$ s. At $1/\lambda_M = 0.06$ s the mixing already keeps the identification probability close to the random guess, and the jitter changes little.

The jitter influences the anonymity through η in the same way as the mixing delay, because it enters the noise $\sigma_{\tilde{Z}}$ of equation (4.18). Indeed, configurations with the same η obtained from different combinations of mixing delay and jitter have the same identification probability within their uncertainty. This suggests that the noise of the network substitutes

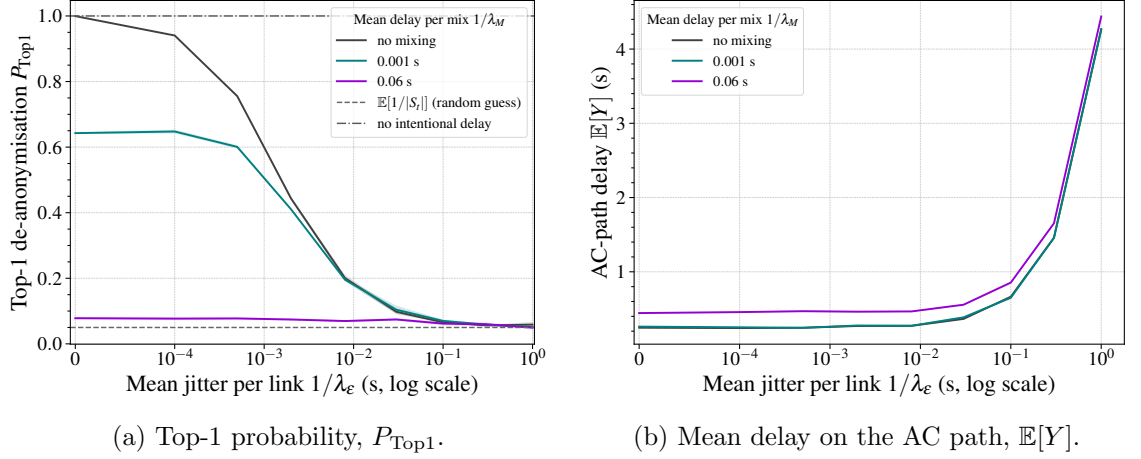


Figure 5.9: (a) The probability that the sender is named and (b) the mean delay on the AC path against the mean jitter per link for the single-path mix at $L = 3$, one curve per mean delay per mix, with cover fixed at 19 senders per slot. Lines are the simulation mean over 30 epochs. In (a) the dashed horizontal line is the random-guess probability and the dash-dotted line the limit of no delay.

for the mixing delay, and a designer whose network is subject to jitter may reduce the mixing delay by the same amount of noise. However, the jitter adds latency of its own, as Figure 5.9b shows. It delays every link of the AC path and of the broadcast, and the designer does not control its magnitude.

Cover is the last parameter, and the only one that changes the candidate set rather than the timing. It is drawn with probability p_s on the single path at three mean delays per mix. Figure 5.10 shows P_{Top1} against p_s , with the bandwidth overhead.

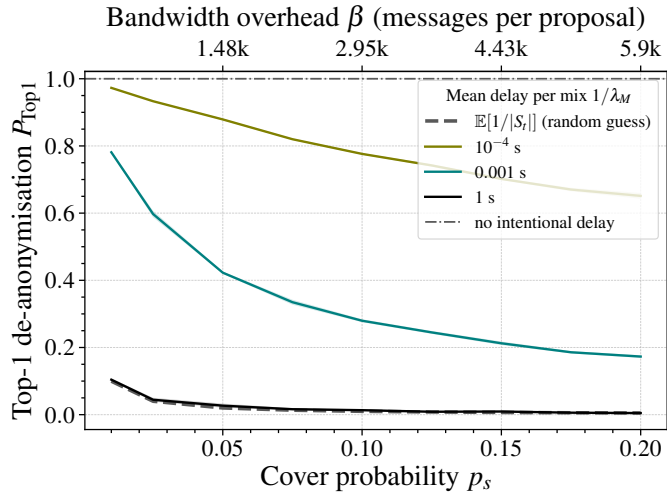


Figure 5.10: The probability that the sender is named against the cover probability for the single-path mix at $L = 3$, one curve per mean delay per mix. Lines are the simulation mean over 30 epochs. The grey dashed curve is the random-guess probability at each p_s , and the dash-dotted line is the limit of no intentional delay. The upper axis shows the bandwidth overhead β of each p_s .

the random-guess probability itself. At the smaller delays, the excess over the random guess decreases with p_s , but slowly.

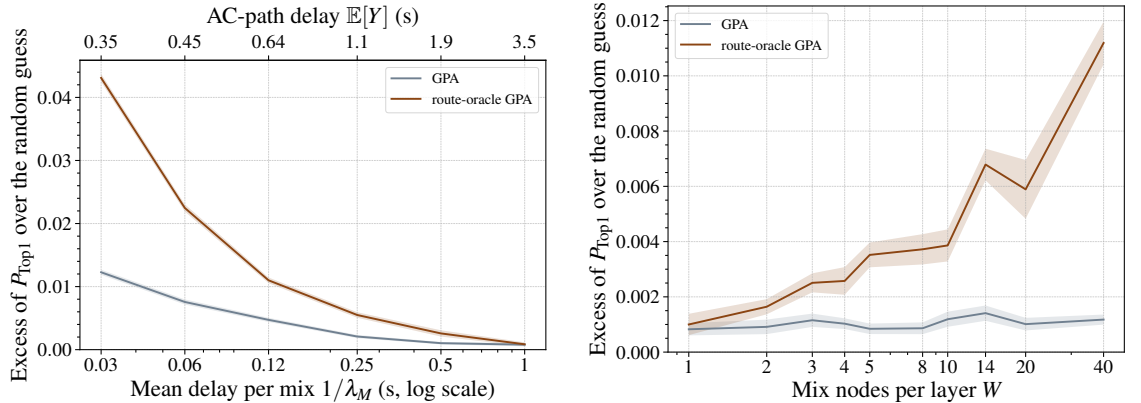
Cover is the most expensive of the parameters. Its bandwidth overhead increases with p_s , whereas the delay costs no bandwidth, and for the same anonymity cover costs far more bandwidth than the delay costs latency. Hence, against sender attribution the delay is the parameter of choice, and cover is reserved for the case in which latency cannot be afforded. Its value is set by stake inference, for which it is the only defence.

5.2.1 RECOMMENDATION AND THE ROUTE-ORACLE BOUND

The recommended system retains $p_s = 0.1$ and $k = 3$ from stake inference, together with $\rho = 4$ and the split rule. The depth, the mean delay per mix, and the width remain to be fixed.

The depth is fixed first. As Figure 5.5 shows, depth adds latency without adding anonymity that a longer delay at fewer mixes does not, and its value lies in the tolerance to compromised mix nodes, which should improve with L . Loopix and Nym use three layers (Piotrowska et al., 2017; Diaz et al., 2021), and a smaller depth weakens that tolerance whereas a larger one adds latency. Thus, $L = 3$ is retained.

The mean delay per mix is then fixed. In Figure 5.11a, the excess of both adversaries decreases with $1/\lambda_M$. From $1/\lambda_M = 0.5$ s, each adversary lies close to its own random guess. The same convergence is found at every width from 1 to 40 (not shown), and the delay can therefore be fixed before the width. Doubling $1/\lambda_M$ to 1 s reduces the remaining excess further but nearly doubles the delay on the AC path. Thus, $1/\lambda_M = 0.5$ s is recommended. The width is fixed last. Figure 5.11b separates the two adversaries. At



(a) Excess over the random guess against the mean delay per mix at $W = 4$.

(b) Excess over the random guess against the width at $1/\lambda_M = 0.5$ s.

Figure 5.11: The excess of the probability that the sender is named over the random guess of each adversary, (a) against the mean delay per mix at $W = 4$ and (b) against the width W at a mean delay per mix of 0.5 s, for the stratified mix network at $L = 3$ under the split rule with $p_s = 0.1$ and $k = 3$, for the GPA and the route-oracle GPA. Lines are the simulation mean over 30 epochs with a band of one SEM.

$1/\lambda_M = 0.5$ s the excess of the GPA shows no trend over W , whereas that of the route-oracle GPA increases overall. The difference arises from the candidate set. Under the split rule, the route reveals the entry node of the sender, and the candidates of the route-oracle GPA are the senders attached to that node, a subset of S_t smaller by a factor of about W . This suggests that the same timing evidence raises the identification probability by more when the true sender is one of fewer candidates, whereas the candidate set of the GPA is S_t regardless of W . Hence, at the recommended delay, width adds no anonymity against the GPA and reduces it against the route-oracle GPA.

The choice of W is therefore a trade-off between capacity and the anonymity against an adversary that learns the route, which Figure 5.11b quantifies. The loss increases with W without a threshold. A moderate width is taken here, at which the excess of the route-oracle GPA is still of the order of that of the GPA, whereas at the width of the deployed Nym network it is an order of magnitude larger. Thus, $W = 4$ is recommended.

The route-oracle GPA is, in effect, the adversary of a single-path mix, as a message whose route is known follows one path of nodes. The same construction applies to any AC system in which messages follow paths of nodes. Given the path of every message, the system reduces to a single-path mix with the candidate set restricted by that path, and the exact posterior of that adversary is an upper bound on the probability that the sender is linked to a message. This suggests a general method for an upper bound on sender linkability, in which the path is granted rather than inferred. Two caveats apply. The bound concerns the path only, as the delays remain unobserved, and it holds per proposal, because the adversary does not combine evidence across proposals.

5.2.2 COMPUTATIONAL COST OF THE ATTACK

The GPA is assumed to have unlimited computation. In the stratified mix network this assumption requires justification. The likelihood of equation (4.12) sums W^{L-1} terms for every candidate, and a sufficiently wide and deep network might place the sum beyond any adversary. Two algorithms are compared, the enumeration of the routes used by the attack above and a dynamic programming (DP) algorithm that returns the same posterior in polynomial time, up to the resolution of a grid.

The unit of work is one evaluation of the density $f_{\tilde{Z}}$ of equation (4.9) for one candidate and one route. Over an epoch the enumeration performs

$$K W^{L-1} \text{ evaluations, where } K = \sum_{b=1}^B |S_{t_b}| \quad (5.7)$$

is the total number of candidates of its B proposals. With cover fixed at 19 senders per slot, $K \approx B \mathbb{E}[|S_t|] = T\lambda(20 + \lambda) \approx 264,000$, against a measured 264,092. The running time per epoch is approximately $K W^{L-1} \tau(L)$, where $\tau(L)$ is the measured cost of one

evaluation (see Appendix A.5). This cost increases linearly with L , because equation (4.8) has L terms, and it is about 50 ns at $L = 3$ on one core of a laptop processor.

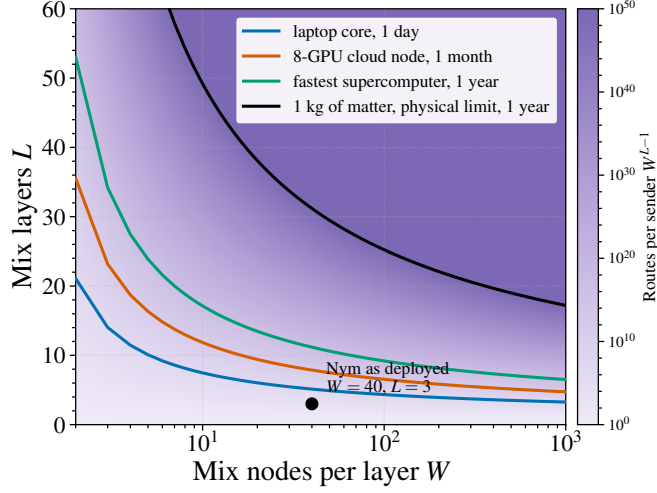
Figure 5.12a inverts the model. For each width it shows the largest depth whose enumeration completes within a given computation. The four levels run from one core of a laptop processor for one day to the physical limit of one kilogram of matter for one year (Lloyd, 2000). The deployed Nym network, with $W = 40$ and $L = 3$ (Diaz et al., 2021), lies below every curve. The depth is the exponent of the route count and the width only its base, and hence one additional layer multiplies the cost by W , whereas a wider network raises it only polynomially. The exponential belongs to the enumeration rather than to the problem. The latency $D_M(\pi)$ of a route is a sum of one link per layer. The sum over routes in equation (4.12) can therefore be computed layer by layer, like the partition function of a chain as a product of transfer matrices (Kramers and Wannier, 1941), an instance of dynamic programming (Bellman, 1966). The distribution of the accumulated latency is propagated from the entry node through the $L - 1$ inter-layer steps on a grid of G values of the delay and convolved once with $f_{\tilde{Z}}$. The likelihood of each candidate is then read from the resulting table, once per exit node. The cost is $O(LW^3G)$ for the propagation, one pass per entry node under the split rule, and $O(KW)$ for the candidates, against $O(KW^{L-1})$ for the enumeration. Both algorithms use the same knowledge, and their posteriors agree to the resolution of the grid. Figures 5.12b and 5.12c show the consequence. At a fixed width, the running time of the enumeration increases exponentially with the depth, whereas at a fixed depth it increases only as W^2 and is below one day up to $W = 1000$. The DP remains below the enumeration at every width and, from $L = 3$, at every depth. Hence, the depth determines the cost of the enumeration, the width enters only polynomially, and the DP reduces the dependence on the depth to a linear one.

Consequently, the computation assumed of the GPA is available to any adversary with polynomially bounded resources, and neither the width nor the depth of the network protects by intractability. The protection measured in this chapter, the noise and the routes, is information-theoretic and survives an adversary with unlimited computation. Thus, the depth is bounded by the latency limit of equation (3.22) and not by the cost of the attack.

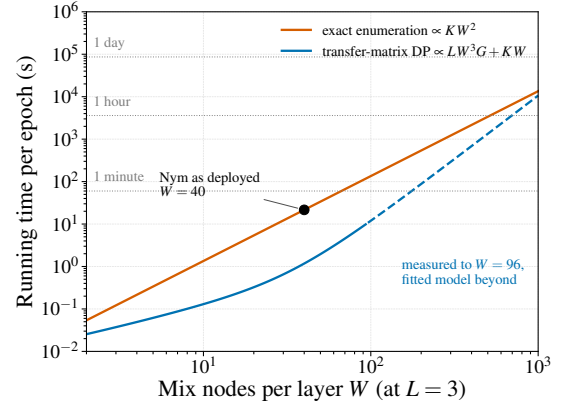
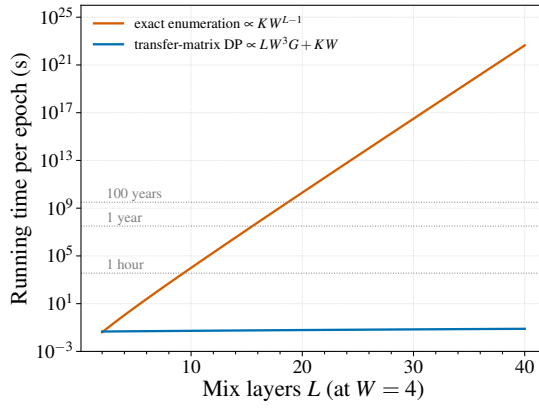
5.3 THE RECOMMENDED SYSTEM

The parameters fixed by the two attacks form one AC system. Table 5.1 lists its parameters, costs, and residual identification probabilities, and Figure 5.13 shows the system. The stake limit is a requirement on the chain rather than a parameter of the AC system, and the link jitter is set to zero as the case that favours the adversary.

Figure 5.14 shows the two costs of the recommended system. The mean propagation



(a) The largest depth whose enumeration completes, for four levels of computation.



(b) Running time against the depth at $W = 4$. (c) Running time against the width at $L = 3$.

Figure 5.12: (a) The largest depth L at each width W for which the enumeration of the routes completes within the available computation, for one core of a laptop processor for one day, a cloud node with eight graphics processing units (GPUs) for one month, the fastest supercomputer for one year (TOP500, 2025), and the physical limit of one kilogram of matter for one year (Lloyd, 2000), over the number of routes per sender W^{L-1} (shading). (b) and (c) The running time of the attack for one epoch by enumeration and by the transfer-matrix DP, (b) against the depth at $W = 4$ and (c) against the width at $L = 3$, where the DP is measured up to $W = 96$ and extrapolated beyond (dashed). The marker is the deployed Nym network, $W = 40$ and $L = 3$. All times are for that core, with cover fixed at 19 senders per slot and $K = 264,092$.

delay of a block is 2.1 s, 7% of the limit of equation (3.22), so that the mean delay per mix can still be increased should a stronger adversary require it. The bandwidth overhead of 2,950 messages per proposal is the more demanding of the two costs. Whether this bandwidth is affordable is a decision for the designer, and a lower p_s reduces it in proportion, at the loss of stake privacy shown in Figure 5.3.

Of the two results, only sender attribution transfers unchanged to a chain with the Cardano-calibrated stake distribution, because it does not depend on the stake distribu-

Parameters			
Parameter	Symbol		Value
Stake limit per validator	$\mathbb{E}[M_N]$	about 1%, i.e. $k \geq 3$	
Cover probability	p_s		0.1
Depth	L		3
Mean delay per mix	$1/\lambda_M$		0.5 s
Sender delay ratio	ρ	4, i.e. $1/\lambda_S = 0.125$ s	
Width	W		4
Entry rule			split
Link jitter	$1/\lambda_\varepsilon$		0
Costs			
Cost	Symbol		Value
Bandwidth overhead	β	2,950 messages per proposal	
Mean delay on the AC path	$\mathbb{E}[Y]$		1.9 s
Mean propagation delay	$\mathbb{E}[\Delta]$	2.1 s, 7% of the limit	
Latency overhead	ℓ		8.5
Load per mix node	$p_s N/W$	25 messages per slot	
Residuals			
Residual	Symbol		Value
Largest stakeholder identified, per epoch	$P_{\max}$	0.010, against $1/N = 0.001$	
Sender named, GPA	P_{Top1}	0.011, against 0.010	
Sender named, route-oracle GPA	P_{Top1}	0.043, against 0.040	

Table 5.1: The recommended AC system for $N = 1000$ validators, $f = 1/30$, and $T = 388,800$ slots, its costs, and the residual identification probabilities at the recommended values. Costs and residuals are the simulation mean.

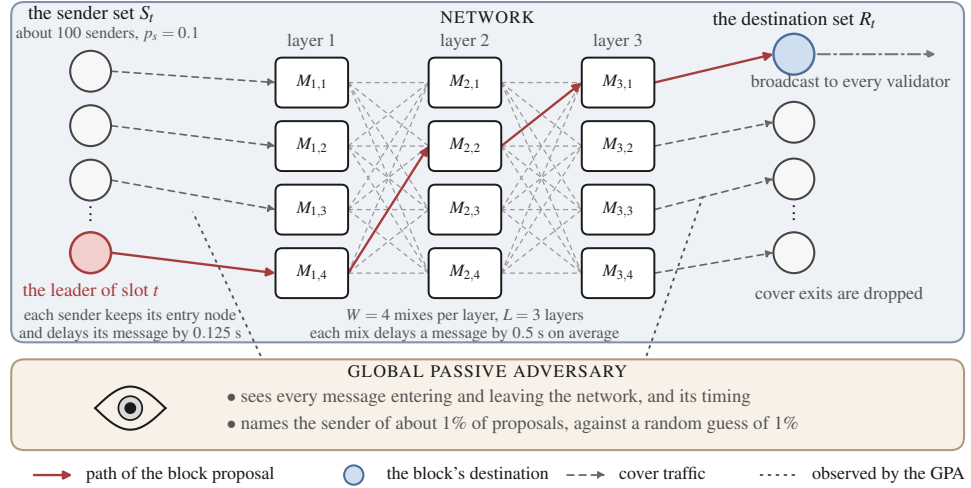
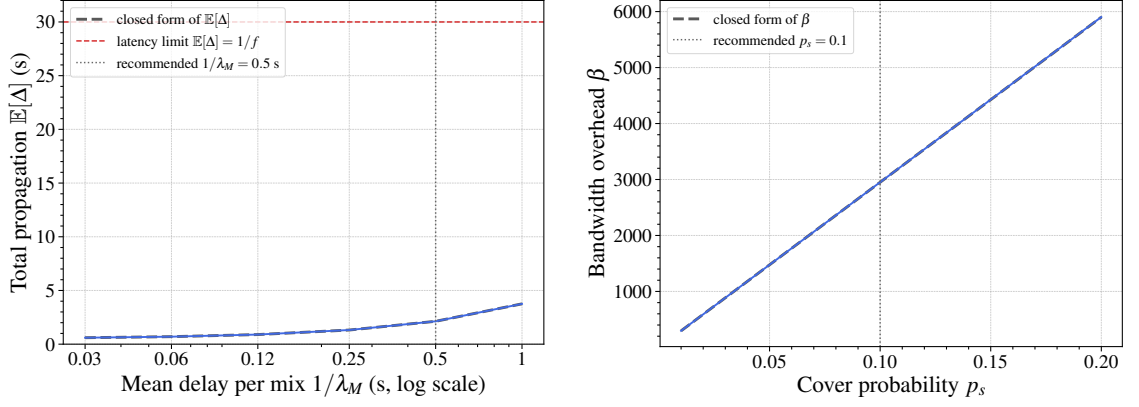


Figure 5.13: The recommended AC system. The stratified mix network has $L = 3$ layers of $W = 4$ mixes, each delaying a message by 0.5 s on average. Each sender keeps one entry node, delays its own message by 0.125 s on average, and sends cover with probability $p_s = 0.1$.

tion. Without the stake limit, the recommendation therefore protects the sender of a proposal but not the identity of the largest stakeholder.



(a) Mean propagation delay against the mean delay per mix. (b) Bandwidth overhead against the cover probability.

Figure 5.14: (a) The mean propagation delay $\mathbb{E}[\Delta]$ of a block under the recommended system against the mean delay per mix, as the simulation mean over 30 epochs at $W = 4$ and the closed form of equation (3.21), with the latency limit of equation (3.22). (b) The bandwidth overhead β against the cover probability, as the simulation mean over 120 stake vectors of 25 epochs each with a band of one standard deviation, and the closed form of equation (3.23). Dotted lines mark the recommended values.

5.4 NETWORK SIZE

The results above are for a network of $N = 1000$ validators, whereas a deployed chain has a different and changing number of validators. The recommended system is therefore evaluated at $N = 100$ and $N = 3000$. The larger size is of the order of the number of stake pools on Cardano, about 2,700 with active stake (Koios, 2026), and the smaller one represents a small chain.

Figure 5.15 shows the excess of P_{Top1} over the random guess of the GPA against N . The random guess decreases as the sender set increases with the network, and the excess decreases in the same proportion. Hence, the size of the network enters only through the sender set. In absolute terms the larger chain is the more anonymous, and the GPA is strongest on the small chain.

Figure 5.16 shows P_{max} and J_x against N at the two shapes. Both measures decrease with N at both shapes, because the expected maximum relative stake of equation (5.3) decreases as $N^{1/k-1}$. The largest stakeholder of a larger network holds a smaller share, wins fewer slots, and its count is exceeded by chance by more of the other validators. At $k = 3$ the largest stakeholder is still identifiable on the small chain, whereas at $N = 3000$ the probability is indistinguishable from the random guess. At $k = 4/3$ the expected maximum share decreases much less with N , and the largest stakeholder of the Cardano-calibrated chain remains identifiable at every size. The stake limit therefore matters more than the size of the network.

Figure 5.17 shows the two costs against N . The latency of the AC path does not depend

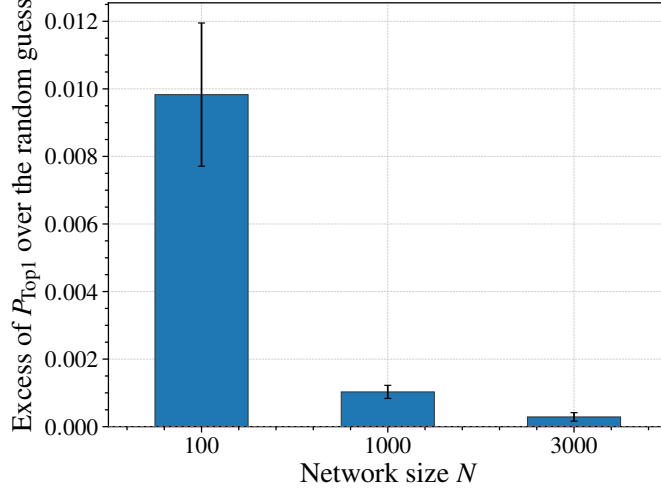


Figure 5.15: The excess of the probability that the sender is named over the random guess $\mathbb{E}[1/|S_t|]$ against the network size N , for the GPA on the recommended stratified mix network with $W = 4$, $L = 3$, a mean delay per mix of 0.5 s, $p_s = 0.1$, and $k = 3$. Bars are the simulation mean over 30 epochs with error bars of one SEM.

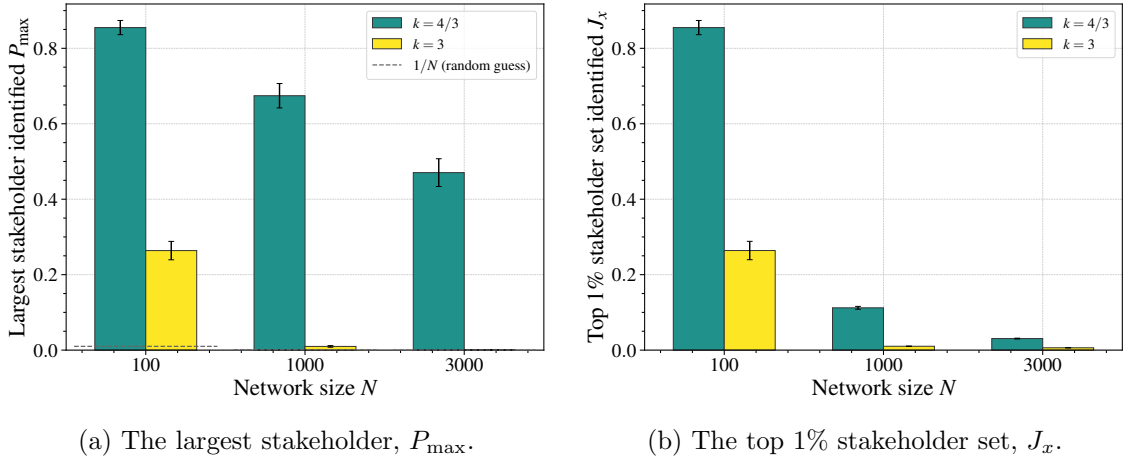
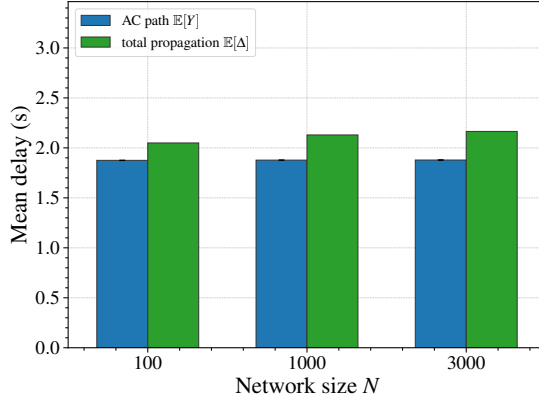
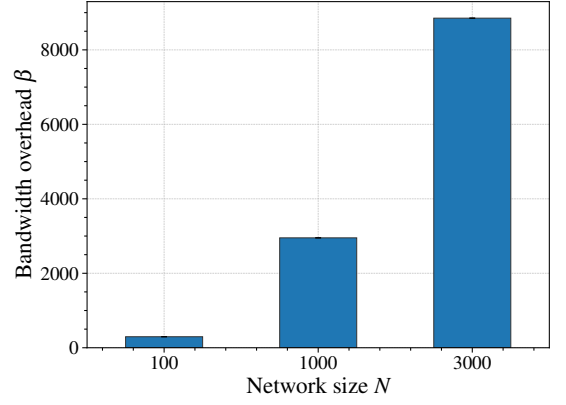


Figure 5.16: Both measures against the network size N at $p_s = 0.1$ for the Cardano-calibrated shape $k = 4/3$ and the recommended shape $k = 3$. Bars are the simulation mean over 120 stake vectors of 25 epochs each with error bars of one SEM. The dashed marks in (a) are the random guess $1/N$. The top 1% set comprises 1, 10, and 30 validators, so that (b) coincides with (a) at $N = 100$.

on the size of the network, as the path comprises a fixed number of links and delays, and the mean propagation delay $\mathbb{E}[\Delta]$ increases only through the broadcast latency $\mathbb{E}[D_{\text{br}}]$, which scales as $\log N$ with the distances of the gossip graph. The cost of size lies in bandwidth. At a fixed p_s , the overhead β of equation (3.23) increases in proportion to N , as cover is sent per validator whereas block production is fixed by f . Consequently, growth changes the sender set and the aggregate bandwidth, and it leaves the residual relative to the random guess, the latency, and the cost per validator unchanged.



(a) The mean delay on the AC path and the mean propagation delay.



(b) The bandwidth overhead, β .

Figure 5.17: (a) The mean delay on the AC path $\mathbb{E}[Y]$ and the mean propagation delay $\mathbb{E}[\Delta] = \mathbb{E}[Y] + \mathbb{E}[D_{\text{br}}]$, and (b) the bandwidth overhead β , against the network size N for the recommended system at $p_s = 0.1$. Bars in (a) are the simulation mean over 30 epochs and in (b) over 120 stake vectors of 25 epochs each.

CHAPTER 6

CONCLUSIONS

The traffic of a PPoS blockchain exposes its leaders unless it is protected by an AC system. Without one, the GPA identifies the leaders and recovers the distribution of stake from the traffic it observes.

Stake inference is governed by the concentration of stake rather than by the AC system. At the Cardano-calibrated concentration, the largest stakeholder is identified within weeks, whereas a limit on the stake per validator reduces the identification to a small residual per epoch and postpones it to years. Hence, the stake limit is a requirement on the chain. Cover lowers the residual only gradually, and each further unit of bandwidth yields less.

Bayesian sender attribution is governed by the ratio of the latency differences between the candidates to the random delay. The mean delay per mix is therefore the principal defence, and the depth and the jitter of the network enter through the same ratio. The parameters of the AC system occupy different positions on the anonymity trilemma. The delay obtains anonymity at a cost in latency, and cover reduces the random guess itself at a cost in bandwidth, whereas width costs neither and conceals only where the delay is small. Neither the width nor the depth protects by intractability, as the attack runs in polynomial time. A stratified mix network of moderate width and delay keeps both adversaries close to their random guesses at a small fraction of the latency limit of the protocol. The size of the network changes only the sender set and the aggregate bandwidth.

The hypothesis of this study is supported in both of its parts. The traffic induced by a PPoS blockchain exposes the metadata from which the GPA recovers the leaders and the distribution of stake, and a mix-based AC system suppresses this leakage to a small and quantified residual. The qualification concerns the cost. For sender attribution, the residual is obtained at a latency within the limit of the consensus protocol. For stake privacy, it requires the stake limit on the chain and a bandwidth overhead that the designer must judge affordable, and it accumulates under observation over years.

6.1 STRENGTHS AND LIMITATIONS

The components of the simulation are validated against closed-form results, and the stake measures and the two costs are reported beside their closed forms. The attribution attack is an exact posterior, whose leakage is not limited by any approximation, and two adversaries bound the effect of the route that the GPA does not observe. The magnitudes are calibrated rather than assumed, specifically the stake distribution to Cardano, the link latencies to measured internet pings, and the protocol constants to the Logos Blockchain. The framework is also modular and seeded, and every experiment covers a full epoch. Consequently, every reported result is reproducible.

The limitations concern the adversary, the model, and the scope. The attack uses the entry and the exit of the AC path but not the observations of the links between the mixes. Observation at every mix node would add the times at which messages cross those links. Part of this evidence, the route, is bounded by the route-oracle GPA, and the remainder is not measured. Perhaps the entry and exit observations already capture most of the timing evidence, but this is a hypothesis rather than a result. Further assumptions favour the adversary. The GPA knows every quenched link latency, and it is assumed to know the slot in which every observed message was sent, although the sender delays could make the messages of consecutive slots overlap in time. Both are worst-case assumptions, as is the split rule, and the leakage reported is an upper bound on what an adversary with less knowledge would obtain. In contrast, the entry is recorded at slot resolution, which withholds the sender delay from the GPA and favours the defence, although the share of that delay in the variance of the random delay is small. The inference, however, is one procedure, as the posterior is computed per proposal and never combines evidence across proposals. The results therefore hold against one inference by an adversary with worst-case knowledge. A less informed adversary would extract less, and a sharper inference could extract more.

The model simplifies the deployed systems in several respects. The mixes are external to the gossip network, and the mixing delays do not depend on load. Hence, cover adds no latency by construction, and the tracing of a route through a sparsely loaded mix is bounded by the route-oracle GPA rather than modelled. Cover is slot-synchronous rather than continuous in time, the mixes emit no loop cover, and the gateway layer is represented by the fixed attachment of validators to entry nodes. The stake distribution is a single Pareto shape, which reproduces the top share of Cardano but not its overall inequality, and the stake vector is fixed across epochs.

The scope is also limited. The recommendation optimises anonymity, latency, and bandwidth and says nothing about availability. Notably, a stratified network is disconnected by the loss of every node of one layer, and its tolerance to node loss, like that to compromised nodes, is taken from the literature rather than measured. Finally, the re-

sults are those of a simulation on one machine rather than of a deployment, at $N = 1000$ validators with two further sizes.

6.2 FUTURE WORK

The most direct extension of the attack is the aggregation of evidence across proposals. The per-proposal posteriors of an epoch are the input of an SDA (Danezis, 2003; Mathewson and Dingledine, 2004), which would test whether cover that hides a sender per proposal still hides it over thousands of proposals. The trace already contains every observation that such an attack needs, and the result would state the time to identification for sender attribution as the many-epoch curves state it for stake inference. In the same direction, the slot in which a message was sent can be treated as a latent variable of the likelihood and marginalised as the route is (Troncoso and Danezis, 2009), which would remove one worst-case assumption and measure its effect. A measure of unobservability, i.e. whether the GPA can distinguish a slot with a proposal from one without, would quantify the benefit of the cover sent in empty slots, which the attribution measures do not capture.

Stronger adversaries remain to be evaluated. An adversary that observes every mix node would test whether the entry and exit observations suffice. A corrupted-node variant, in which the adversary observes the interior of some mix nodes without altering their behaviour, corresponds to the threat model of deployed mix networks (Piotrowska et al., 2017; Diaz et al., 2021), whose tolerance to compromised nodes is otherwise assumed here. An active adversary, which delays or drops messages, is the case for which the loop cover of Loopix is designed. In the opposite direction, the exact knowledge of the quenched link latencies can be relaxed to knowledge of the link law, which enters the Bayesian framework as a prior over the latencies. The jitter already has this form, as the GPA knows its rate but not its draw, and this suggests that the relaxation would reduce identification in the same way as additional noise.

The route-oracle construction merits development as a general tool. Applied, for example, to the relay paths of Dandelion++ or the circuits of onion routing (Fanti et al., 2018; Dingledine et al., 2004), it would place different topologies on one scale. The conditions under which the reduction is exact remain to be stated.

On the model side, the internal architecture, load-dependent mixing delays, and continuous-time cover with loop cover from the mixes would bring the model closer to the deployed systems. A two-component Pareto mixture with stake dynamics would do the same for the chain. A protocol-level defence also follows from the results, as the stake limit per validator could be enforced by the protocol rather than assumed. In addition, an availability axis, i.e. the fraction of nodes whose loss disconnects the network, would complete the comparison of topologies.

The dissertation connects the consensus proofs, which assume an anonymous broadcast

channel, with the AC systems that would have to provide it, and measures that channel on the traffic it must protect. It reports the defence configurations on the axes of the anonymity trilemma (Das et al., 2018) and adds the stake-privacy question that generic AC studies do not ask. For this setting, it addresses the open problems of Diaz et al. (2002), i.e. anonymity under prolonged observation and the quantitative benefit of cover traffic. It also shows that the size of a mix network offers no computational protection. For practice, the framework allows the designer of an AC system to evaluate a defence before deployment and to re-derive the recommendation for a different latency or bandwidth allowance, stake distribution, or adversary. The immediate application is the Blend Network of the Logos Blockchain (Logos, 2026b). The results also inform the protocol designer, for whom the stake limit per validator is the requirement without which no AC system of the kind evaluated here protects the largest stakeholder. Beyond the designers, they serve the users of such chains, for whom the exposure of a leader is a real threat.

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APPENDIX A

DERIVATIONS

A.1 LEADER ELECTION STATISTICS

The number of leaders in slot t is $n_t = \sum_{i=1}^N s_i(t)$, where the outcome $\mathbf{s}(t)$ of the slot follows the product law of equation (3.2), whose factor for validator i is denoted $\mathbb{P}_i(s_i(t)) = \phi(\alpha_i)^{s_i(t)}[1 - \phi(\alpha_i)]^{1-s_i(t)}$. The average, the variance, and the distribution of n_t are derived from this law.

The average of n_t is computed as follows:

$$\begin{aligned}
 \mathbb{E}[n_t] &= \sum_{\mathbf{s}(t)} \mathbb{P}(\mathbf{s}(t)) \sum_{i=1}^N s_i(t) \\
 &= \sum_{i=1}^N \sum_{\mathbf{s}(t)} \mathbb{P}(\mathbf{s}(t)) s_i(t) \\
 &= \sum_{i=1}^N \sum_{s_i(t)} \mathbb{P}_i(s_i(t)) s_i(t) \\
 &= \sum_{i=1}^N \phi(\alpha_i).
 \end{aligned} \tag{A.1}$$

Furthermore, it follows that the average number of leaders per epoch is $T \sum_{i=1}^N \phi(\alpha_i)$. Consider the sum $\sum_{i=1}^N \phi(\alpha_i) = N \sum_{i=1}^N \frac{1}{N} \phi(\alpha_i)$. The function ϕ is concave, and by Jensen's inequality

$$\sum_{i=1}^N \phi(\alpha_i) \leq N \phi\left(\frac{1}{N} \sum_{i=1}^N \alpha_i\right) = N \phi(1/N), \tag{A.2}$$

with equality for $\alpha_i = 1/N$. Moreover, for N large,

$$N \phi(1/N) = -\ln(1 - f) - \frac{\ln^2(1 - f)}{2N} + O\left(\frac{1}{N^2}\right). \tag{A.3}$$

In general, for α_i small, the expansion $\phi(\alpha_i) = -\ln(1-f)\alpha_i - \ln^2(1-f)\alpha_i^2/2 + O(\alpha_i^3)$ and $\sum_{i=1}^N \alpha_i = 1$ yield

$$\begin{aligned}\sum_{i=1}^N \phi(\alpha_i) &= \sum_{i=1}^N \left[-\ln(1-f)\alpha_i - \frac{\ln^2(1-f)\alpha_i^2}{2} + O(\alpha_i^3) \right] \\ &= -\ln(1-f) - \frac{\ln^2(1-f)}{2} \sum_{i=1}^N \alpha_i^2 + O\left(\sum_{i=1}^N \alpha_i^3\right),\end{aligned}\quad (\text{A.4})$$

which is equation (3.4). Notably, $\sum_{i=1}^N \alpha_i^p < \sum_{i=1}^N \alpha_i = 1$ for $p > 1$.

The variance is

$$\text{Var}(n_t) = \mathbb{E}[n_t^2] - \mathbb{E}[n_t]^2, \quad (\text{A.5})$$

which can be computed as follows. First, the second moment is

$$\mathbb{E}[n_t^2] = \sum_{i=1}^N \sum_{j=1}^N \sum_{\mathbf{s}(t)} \mathbb{P}(\mathbf{s}(t)) s_i(t) s_j(t) = \sum_{i=1}^N \sum_{j=1}^N \mathbb{E}[s_i(t) s_j(t)]. \quad (\text{A.6})$$

Now the average $\mathbb{E}[s_i(t) s_j(t)]$ equals $\phi(\alpha_i)$ for $i = j$, because $s_i(t)^2 = s_i(t)$, and $\phi(\alpha_i)\phi(\alpha_j)$ for $i \neq j$, by independence. Hence,

$$\begin{aligned}\mathbb{E}[n_t^2] &= \sum_{i=1}^N \phi(\alpha_i) + \sum_{i \neq j} \phi(\alpha_i) \phi(\alpha_j) \\ &= \sum_{i=1}^N \phi(\alpha_i) + \left(\sum_{i=1}^N \phi(\alpha_i) \right)^2 - \sum_{i=1}^N \phi^2(\alpha_i).\end{aligned}\quad (\text{A.7})$$

Second, using this result and the average of equation (A.1) in the definition of equation (A.5) yields

$$\text{Var}(n_t) = \sum_{i=1}^N [\phi(\alpha_i) - \phi^2(\alpha_i)] = \sum_{i=1}^N \phi(\alpha_i) [1 - \phi(\alpha_i)], \quad (\text{A.8})$$

which is the variance of equation (3.3). Equivalently, $\text{Var}(n_t) = \mathbb{E}[n_t] - \sum_{i=1}^N \phi^2(\alpha_i)$.

The distribution of n_t is

$$\begin{aligned}\mathbb{P}(n_t = n) &= \sum_{\mathbf{s}(t)} \mathbb{P}(\mathbf{s}(t)) \mathbf{1}\left[\sum_{i=1}^N s_i(t) = n\right] \\ &= \sum_{\mathbf{s}(t)} \prod_{i=1}^N \phi(\alpha_i)^{s_i(t)} [1 - \phi(\alpha_i)]^{1-s_i(t)} \mathbf{1}\left[\sum_{i=1}^N s_i(t) = n\right].\end{aligned}\quad (\text{A.9})$$

This law defines the probability $\mathbb{P}(n_t \geq n_0) = 1 - \sum_{n=0}^{n_0-1} \mathbb{P}(n_t = n)$, in which $\mathbb{P}(n_t < n_0)$

is the sum over the outcomes with fewer than n_0 leaders:

$$\begin{aligned}\mathbb{P}(n_t < n_0) &= \prod_{i=1}^N [1 - \phi(\alpha_i)] + \sum_{i=1}^N \phi(\alpha_i) \prod_{j \neq i} [1 - \phi(\alpha_j)] \\ &\quad + \sum_{i_1 < i_2} \phi(\alpha_{i_1}) \phi(\alpha_{i_2}) \prod_{j \notin \{i_1, i_2\}} [1 - \phi(\alpha_j)] + \dots \\ &\quad + \sum_{i_1 < \dots < i_{n_0-1}} \phi(\alpha_{i_1}) \dots \phi(\alpha_{i_{n_0-1}}) \prod_{j \notin \{i_1, \dots, i_{n_0-1}\}} [1 - \phi(\alpha_j)].\end{aligned}\quad (\text{A.10})$$

For $n_0 = 1$, the probability that there is at least one leader in slot t is $\mathbb{P}(n_t \geq 1) = 1 - \prod_{i=1}^N [1 - \phi(\alpha_i)] = f$, which is equation (3.5). For $n_0 = 2$,

$$\begin{aligned}\mathbb{P}(n_t < 2) &= \prod_{i=1}^N [1 - \phi(\alpha_i)] + \sum_{i=1}^N \phi(\alpha_i) \prod_{j \neq i} [1 - \phi(\alpha_j)] \\ &= \left[1 + \sum_{i=1}^N \frac{\phi(\alpha_i)}{1 - \phi(\alpha_i)} \right] \prod_{j=1}^N [1 - \phi(\alpha_j)] \\ &= (1 - f) \left[1 + \sum_{i=1}^N \frac{\phi(\alpha_i)}{1 - \phi(\alpha_i)} \right],\end{aligned}\quad (\text{A.11})$$

and hence

$$\mathbb{P}(n_t \geq 2) = 1 - \mathbb{P}(n_t < 2) = f - (1 - f) \sum_{i=1}^N \frac{\phi(\alpha_i)}{1 - \phi(\alpha_i)}, \quad (\text{A.12})$$

which is equation (3.6).

A.2 THE LATENCY LIMIT OF THE CONSENSUS PROTOCOL

The Logos protocol infers the total active stake from the observed chain growth. An honest validator observes the number U_n of blocks on its chain during epoch n , and the inferred total stake $\hat{\Omega}_n$ is updated towards the target growth rate f by the recursion

$$\hat{\Omega}_{n+1} = \hat{\Omega}_n - \gamma \hat{\Omega}_n \left(f - \frac{U_n}{T} \right), \quad (\text{A.13})$$

where $\gamma > 0$ is the learning rate. Network delays cause forks, of which only one block survives under the longest-chain rule, and U_n therefore falls below the number of produced blocks, i.e. the total stake is underestimated.

Consider first a known total stake $\Omega = \sum_{i=1}^N \omega_i$. By equation (3.5), a slot produces a block with probability f . A block produced in slot s reaches every validator after the propagation delay Δ_s , and a leader of a later slot that has not received it builds on the previous block instead. Let $\Delta_s^+ = \lfloor \Delta_s \rfloor + 1$ denote the separation in slots after which the block of slot s has been received, which equals 1 without delay. Two surviving blocks in

slots $s < t$ must therefore satisfy $t - s \geq \Delta_s^+$. The fraction q of produced blocks that survive follows from this spacing. After a surviving block in slot s , every block produced in the following $\Delta_s^+ - 1$ slots is discarded, and the next surviving block is the first block produced from slot $s + \Delta_s^+$ onwards, which occurs after a geometrically distributed number of slots with mean $1/f$. Consecutive surviving blocks are thus separated by $\mathbb{E}[\Delta^+] - 1 + 1/f$ slots on average, against $1/f$ for produced blocks, and for a general random network delay

$$q = \frac{1/f}{\mathbb{E}[\Delta^+] - 1 + 1/f} = \frac{1}{1 + f(\mathbb{E}[\Delta^+] - 1)}. \quad (\text{A.14})$$

Without delay, $\mathbb{E}[\Delta^+] = 1$ and $q = 1$, and q decreases as $\mathbb{E}[\Delta^+]$ increases.

When the total stake is inferred, network delays create a feedback loop. A delay lowers U_n/T , and by equation (A.13) the inferred stake decreases. The leader election depends on the inferred denominator, with $\phi_i = 1 - (1 - f)^{\omega_i/\hat{\Omega}}$ in place of equation (3.1), and by the product argument of equation (3.5) a slot then produces a block with probability

$$\Phi = 1 - (1 - f)^{\Omega/\hat{\Omega}}, \quad (\text{A.15})$$

which equals f for $\hat{\Omega} = \Omega$ and increases as $\hat{\Omega}$ decreases. With Φ in place of f , the surviving fraction of equation (A.14) is $q = 1/[1 + \Phi(\mathbb{E}[\Delta^+] - 1)]$, and the observed chain-growth rate is $q\Phi$. The recursion is stationary when this rate equals the target, $q\Phi = f$, i.e.

$$\begin{aligned} \frac{\Phi}{1 + \Phi(\mathbb{E}[\Delta^+] - 1)} &= f, \\ \Phi [1 - f(\mathbb{E}[\Delta^+] - 1)] &= f, \end{aligned} \quad (\text{A.16})$$

and at equilibrium

$$\Phi = \frac{f}{1 - f(\mathbb{E}[\Delta^+] - 1)} \quad \text{and} \quad q = \frac{f}{\Phi} = 1 - f(\mathbb{E}[\Delta^+] - 1). \quad (\text{A.17})$$

The protocol compensates for block losses by increasing the pre-fork block production. A positive equilibrium exists only if $\Phi < 1$, which by equation (A.17) requires

$$\mathbb{E}[\Delta^+] < \frac{1}{f}. \quad (\text{A.18})$$

At the critical point $\mathbb{E}[\Delta^+] = 1/f$, block production reaches one block per slot, $\Phi = 1$, and by equation (A.15) the inferred total stake $\hat{\Omega}/\Omega = \ln(1 - f)/\ln(1 - \Phi)$ becomes zero.

The separation $\Delta^+ = \lfloor \Delta \rfloor + 1$ lies between Δ and $\Delta + 1$, so that equation (A.18) bounds the mean propagation delay by $1/f$ to within one slot. Equation (3.22) states this limit as the mean block interval $1/f = 30$ s. It constrains the mean delay rather than its maximum, because only $\mathbb{E}[\Delta^+]$ enters equations (A.14) and (A.17).

A.3 THE DENSITY OF THE INTENTIONAL RANDOM DELAY

The link jitter enters the attack through the convolution of equation (4.9), so that the only random delays considered here are the intentional delays introduced by the AC system. The sender delays every message by $X_S \sim \text{Exp}(\lambda_S)$, while every mix introduces an independent delay $X_{M,j} \sim \text{Exp}(\lambda_M)$ for $j = 1, \dots, L$. The total random delay is therefore

$$Z = X_S + \sum_{j=1}^L X_{M,j}, \quad (\text{A.19})$$

as in equation (3.17). The objective is to derive the probability density $f_Z(z)$.

The probability density of any random variable can be written as the expectation of a Dirac delta,

$$f_Z(z) = \mathbb{E} \left[\delta \left(z - X_S - \sum_{j=1}^L X_{M,j} \right) \right]. \quad (\text{A.20})$$

Substituting the exponential densities yields

$$f_Z(z) = \int_0^\infty dx \lambda_S e^{-\lambda_S x} \prod_{j=1}^L \left[\int_0^\infty dx_j \lambda_M e^{-\lambda_M x_j} \right] \delta \left(z - x - \sum_{j=1}^L x_j \right). \quad (\text{A.21})$$

This expression has a clear probabilistic interpretation. The Dirac delta enforces the constraint that the sum of all intentional delays equals the observed random delay z .

The total delay introduced by the mixes is $V = \sum_{j=1}^L X_{M,j}$. Because the $X_{M,j}$ are independent exponential random variables with the same rate, V follows a Gamma distribution with shape L and rate λ_M , with density

$$f_V(v) = \frac{\lambda_M^L}{(L-1)!} v^{L-1} e^{-\lambda_M v} \quad \text{for } v \geq 0. \quad (\text{A.22})$$

Consequently, $Z = X_S + V$, and the multidimensional integral of equation (A.21) reduces to the convolution

$$f_Z(z) = \int_0^z \lambda_S e^{-\lambda_S x} \frac{\lambda_M^L}{(L-1)!} (z-x)^{L-1} e^{-\lambda_M(z-x)} dx \quad \text{for } z \geq 0. \quad (\text{A.23})$$

This is simply the convolution of an exponential distribution with an Erlang distribution.

Factoring out the exponential term yields

$$f_Z(z) = \frac{\lambda_S \lambda_M^L}{(L-1)!} e^{-\lambda_S z} \int_0^z (z-x)^{L-1} e^{-(\lambda_M - \lambda_S)(z-x)} dx. \quad (\text{A.24})$$

The substitution $u = z - x$ transforms the integral into

$$f_Z(z) = \frac{\lambda_S \lambda_M^L}{(L-1)!} e^{-\lambda_S z} \int_0^z u^{L-1} e^{-(\lambda_M - \lambda_S)u} du. \quad (\text{A.25})$$

Let $\nu = \lambda_M - \lambda_S$. Assuming $\lambda_M \neq \lambda_S$, the incomplete Gamma integral satisfies

$$\int_0^z u^{L-1} e^{-\nu u} du = \frac{(L-1)!}{\nu^L} \left[1 - e^{-\nu z} \sum_{j=0}^{L-1} \frac{(\nu z)^j}{j!} \right]. \quad (\text{A.26})$$

This identity follows by induction on L . For $L = 1$ the integral is $(1 - e^{-\nu z})/\nu$, and for $L > 1$ integration by parts yields

$$\int_0^z u^{L-1} e^{-\nu u} du = -\frac{z^{L-1} e^{-\nu z}}{\nu} + \frac{L-1}{\nu} \int_0^z u^{L-2} e^{-\nu u} du, \quad (\text{A.27})$$

so that the identity for $L - 1$ extends the sum by the term $j = L - 1$. Substituting this identity into the convolution yields the explicit density. The probability density of the total intentional delay is therefore

$$f_Z(z) = \frac{\lambda_S \lambda_M^L}{(\lambda_M - \lambda_S)^L} \left[e^{-\lambda_S z} - e^{-\lambda_M z} \sum_{j=0}^{L-1} \frac{((\lambda_M - \lambda_S)z)^j}{j!} \right] \quad \text{for } z \geq 0, \quad (\text{A.28})$$

and $f_Z(z) = 0$ for $z < 0$. This is equation (4.8), whose constant A is the prefactor. The identity holds for either sign of ν , and in the recommended system, in which $\rho = \lambda_S/\lambda_M = 4$ and ν is negative, the prefactor and the bracket change sign together for odd L , and their product remains a positive density.

The closed-form expression derived above assumes $\lambda_M \neq \lambda_S$, because it contains the factor $(\lambda_M - \lambda_S)^{-L}$. Consequently, direct substitution of $\lambda_M = \lambda_S$ produces the indeterminate form $0/0$. Nevertheless, the limit exists and is easily obtained. Suppose $\lambda_M = \lambda_S = \lambda$. The total random delay is then the sum of the $L + 1$ independent and identically distributed variables $X_S, X_{M,1}, \dots, X_{M,L} \sim \text{Exp}(\lambda)$, and the sum of $L + 1$ independent exponential random variables with common rate λ follows a Gamma (Erlang) distribution, $Z \sim \text{Gamma}(L + 1, \lambda)$. Hence,

$$f_Z(z) = \frac{\lambda^{L+1}}{L!} z^L e^{-\lambda z} \quad \text{for } z \geq 0, \quad (\text{A.29})$$

and $f_Z(z) = 0$ for $z < 0$.

This Gamma density is precisely the continuous limit of the general formula. Using the Taylor expansion $e^{-\lambda_M z} = e^{-(\lambda_S + \nu)z} = e^{-\lambda_S z} e^{-\nu z}$ with $e^{-\nu z} = \sum_{n=0}^{\infty} (-\nu z)^n / n!$,

$$e^{-\nu z} \sum_{j=0}^{L-1} \frac{(\nu z)^j}{j!} = 1 - \frac{(\nu z)^L}{L!} + O(\nu^{L+1}), \quad (\text{A.30})$$

which implies

$$e^{-\lambda_S z} - e^{-\lambda_M z} \sum_{j=0}^{L-1} \frac{(\nu z)^j}{j!} = e^{-\lambda z} \frac{(\nu z)^L}{L!} + O(\nu^{L+1}). \quad (\text{A.31})$$

Substituting this expansion into the general expression yields

$$\begin{aligned} f_Z(z) &= \frac{\lambda_S \lambda_M^L}{\nu^L} \left[e^{-\lambda z} \frac{(\nu z)^L}{L!} + O(\nu^{L+1}) \right] \\ &= \lambda_S \lambda_M^L e^{-\lambda z} \frac{z^L}{L!} + O(\nu). \end{aligned} \quad (\text{A.32})$$

Finally, taking the limit $\nu \rightarrow 0$, or equivalently $\lambda_M \rightarrow \lambda_S = \lambda$, yields

$$\lim_{\lambda_M \rightarrow \lambda_S} f_Z(z) = \frac{\lambda^{L+1}}{L!} z^L e^{-\lambda z}, \quad (\text{A.33})$$

which is exactly the Gamma density obtained directly from the sum of $L + 1$ independent exponential random variables. Thus, the equal-rate case is not a separate model but simply the continuous limit of the general expression. If the destination also delayed each message before broadcasting, the same derivation would hold with L replaced by $L + 1$. In the present model the destination does not delay.

A.4 THE EXPECTED MAXIMUM RELATIVE STAKE

Let $\omega_1, \dots, \omega_N$ be independent draws from the Pareto distribution of equation (3.7) with scale $x_m > 0$ and shape $k > 0$, whose survival function is

$$\mathbb{P}(\omega_i > x) = \left(\frac{x_m}{x} \right)^k \quad \text{for } x \geq x_m. \quad (\text{A.34})$$

The relative stake of validator i is $\alpha_i = \omega_i / \sum_{j=1}^N \omega_j$, and the maximum relative stake is

$$M_N = \max_{1 \leq i \leq N} \alpha_i = \frac{\max_i \omega_i}{\sum_{j=1}^N \omega_j}. \quad (\text{A.35})$$

It satisfies the deterministic bounds $1/N \leq M_N \leq 1$, and for $N = 1000$ the lower bound is 0.1%.

The distribution of M_N does not depend on the scale. Writing $\omega_i = x_m v_i$, where v_i has a Pareto distribution of unit scale, yields

$$\alpha_i = \frac{x_m v_i}{\sum_{j=1}^N x_m v_j} = \frac{v_i}{\sum_{j=1}^N v_j}. \quad (\text{A.36})$$

Therefore, both the relative stakes and M_N are independent of x_m , and their distributions depend only on N and k .

For $k > 1$ the Pareto mean is finite,

$$\mathbb{E}[\omega_i] = \frac{k}{k-1} x_m. \quad (\text{A.37})$$

The unnormalised sample maximum follows the extreme-value approximation

$$\max_i \omega_i \approx x_m N^{1/k} Q, \quad (\text{A.38})$$

where Q has the Fréchet distribution

$$\mathbb{P}(Q \leq q) = \exp(-q^{-k}) \quad \text{for } q > 0. \quad (\text{A.39})$$

At the same time, the law of large numbers yields

$$\sum_{i=1}^N \omega_i \approx N \mathbb{E}[\omega_i] = N \frac{k}{k-1} x_m. \quad (\text{A.40})$$

Combining the two approximations yields

$$M_N \approx \frac{k-1}{k} N^{1/k-1} Q. \quad (\text{A.41})$$

This approximation requires $k > 1$, because it uses the finite Pareto mean.

To show that Q has the Fréchet distribution, let $\omega_{(N)} = \max(\omega_1, \dots, \omega_N)$ denote the unnormalised maximum. By independence, its exact finite- N distribution function is

$$\mathbb{P}(\omega_{(N)} \leq x) = \prod_{i=1}^N \mathbb{P}(\omega_i \leq x) = \left[1 - \left(\frac{x_m}{x} \right)^k \right]^N \quad \text{for } x \geq x_m. \quad (\text{A.42})$$

Now introduce the rescaled maximum $Q_N = \omega_{(N)} / (x_m N^{1/k})$. For $q > 0$,

$$\begin{aligned} \mathbb{P}(Q_N \leq q) &= \mathbb{P}(\omega_{(N)} \leq x_m N^{1/k} q) \\ &= \left[1 - \left(\frac{x_m}{x_m N^{1/k} q} \right)^k \right]^N \\ &= \left(1 - \frac{q^{-k}}{N} \right)^N. \end{aligned} \quad (\text{A.43})$$

Using $(1 - c/N)^N \rightarrow e^{-c}$ as $N \rightarrow \infty$ yields

$$\mathbb{P}(Q_N \leq q) \rightarrow \exp(-q^{-k}) \quad \text{for } q > 0, \quad (\text{A.44})$$

which is the distribution function of a unit-scale Fréchet distribution with shape parameter k . Thus, Q_N converges in distribution to Q , and for large N , $\omega_{(N)} \approx x_m N^{1/k} Q$, which is

equation (A.38). The finite- N maximum is not exactly Fréchet distributed. The Fréchet law is its limiting distribution after scaling, and no centring term is required.

By equation (A.41), the approximate distribution function of the maximum relative stake is

$$\mathbb{P}(M_N \leq m) \approx \exp \left[- \left(\frac{\frac{k-1}{k} N^{1/k-1}}{m} \right)^k \right]. \quad (\text{A.45})$$

For $k = 4/3$, the factors are $(k-1)/k = 1/4$ and $1/k - 1 = -1/4$, hence $M_N \approx \frac{1}{4} N^{-1/4} Q$ with $\mathbb{P}(Q \leq q) = \exp(-q^{-4/3})$. For $N = 1000$, $\frac{1}{4} N^{-1/4} \approx 0.04446$. This is an asymptotic approximation, and it does not fully account for the fact that the largest observation is also included in the denominator of equation (A.35).

For a Fréchet variable with shape $k > 1$, the density is $kq^{-k-1} \exp(-q^{-k})$ for $q > 0$, and the substitution $t = q^{-k}$ yields the mean

$$\mathbb{E}[Q] = \int_0^\infty q k q^{-k-1} e^{-q^{-k}} dq = \int_0^\infty t^{-1/k} e^{-t} dt = \Gamma\left(1 - \frac{1}{k}\right). \quad (\text{A.46})$$

Therefore, taking the expectation of equation (A.41) yields

$$\mathbb{E}[M_N] \approx \frac{k-1}{k} N^{1/k-1} \Gamma\left(1 - \frac{1}{k}\right) \quad \text{for } k > 1, \quad (\text{A.47})$$

which is equation (5.3). Its two limits agree with the bounds on M_N . As $k \rightarrow \infty$, the factor $N^{1/k-1}$ tends to $1/N$ and the other two factors to 1. As $k \rightarrow 1$, the Gamma function behaves as $\Gamma(1 - 1/k) \approx k/(k-1)$, which cancels the leading factor, and the expectation tends to 1. For $k = 4/3$, $\mathbb{E}[M_N] \approx \frac{1}{4} N^{-1/4} \Gamma(1/4)$, and for $N = 1000$ this yields $\mathbb{E}[M_{1000}] \approx 0.161$. Thus, the pure extreme-value approximation predicts an average maximum stake of approximately 16.1%.

The large- N formula overestimates the finite- N expectation at $k = 4/3$. A direct Monte Carlo simulation over 2,000,000 stake vectors of size $N = 1000$ yields $\mathbb{E}[M_{1000}] \approx 0.102$, i.e. approximately 10.2%, with a median of 6.5%. The median is substantially below the mean, which reflects the strongly right-skewed distribution. Occasional samples contain one exceptionally large observation that accounts for a major share of the total. Across the shapes of Figure 5.2, the simulated values lie below the large- N approximation, especially when k is close to 1, and as k increases the Pareto tail becomes lighter and the maximum relative stake tends toward the equality limit $1/N$. Convergence to the asymptotic result is slow because the distribution is heavy-tailed.

A.5 CALIBRATION OF THE COMPUTATIONAL COST

The cost of the attack is written as the number of atomic operations multiplied by their measured cost, every constant is calibrated on the implemented attack at the full epoch,

and the model is evaluated where measurement is not possible.

STEP 1. THE UNIT OF WORK. The atomic unit is one evaluation of $f_{\bar{z}}$ for one candidate and one route, i.e. one term of the likelihood of equation (4.12).

STEP 2. THE WORK. The evaluations factorise into candidate rows and routes per row, $K W^{L-1}$ in equation (5.7), with W^{L-1} routes per candidate under the split rule and W^L under the uniform rule. The number K of candidate rows per epoch has a closed form. By equation (3.4), the average number of proposals per epoch is $B = T\lambda = 13,181$. Scoring is per proposal, which size-biases the leader count, as a slot with w leaders contributes w proposals of $19 + w$ candidates each at a cover of 19 senders. Under the Poisson approximation of n_t , the mean number of candidates per proposal is $\mathbb{E}[|S_t|] = 20 + \lambda$, hence

$$K = B \mathbb{E}[|S_t|] = T\lambda(20 + \lambda) = 264,065 \quad (\text{A.48})$$

predicted against 264,092 measured. The cover model enters the cost model only through $\mathbb{E}[|S_t|]$.

STEP 3. FROM WORK TO TIME. The running time is $t = t_0 + K W^{L-1} \tau(L)$, where $\tau(L)$ is the measured cost of one evaluation and t_0 a fixed setup overhead. The depth enters twice, through W^{L-1} evaluations that each cost in proportion to L , because the density of equation (4.8) has L terms, hence

$$\tau(L) = \tau_0 + \tau_1 L \quad \text{nanoseconds per evaluation,} \quad (\text{A.49})$$

where τ_0 is the L -independent cost per element and τ_1 the cost of each additional polynomial term. The constants were obtained by a least-squares fit of the running time of the implemented attack on eight grids with different (W, L) at fixed K , whose regressors $K W^{L-1}$ and $K W^{L-1} L$ separate because the grids vary W and L independently. Both constants are properties of the machine rather than of the algorithm, and the regressors are strongly collinear, so the reportable quantity is τ at a stated depth, 51 ns at $L = 3$ and 144 ns at $L = 10$ on one core of an Intel i7-12800H processor.

STEP 4. FROM WORK TO SPACE. The attack works on matrices with one entry of 8 bytes per candidate and route, streamed in chunks of candidate rows with at most 2^{25} elements resident at a time. The peak memory per element, measured with the chunking disabled in six runs varying (W, L) and the mean delay per mix, increases linearly with the fraction $\mathbb{P}(z_i \geq 0)$ of candidates that survive the elimination of negative residuals, as $8[3.14 + 5.00\mathbb{P}(z_i \geq 0)]$ bytes, i.e. from 45 to 65 bytes as that fraction ranges from about 0.5 to 1, and the worst case of 65 bytes per element is assumed. As the number of resident

elements is capped, the peak memory is

$$65 \text{ B} \times \min(2^{25}, K W^{L-1}) \leq 2.2 \text{ GB} \quad (\text{A.50})$$

for every grid, and memory does not enter the ladder.

STEP 5. INVERTING THE MODEL. An attacker is a machine and a duration, converted to an available computation $t_{\max}$ in core-equivalent seconds through the double-precision peak of the machine as a multiple of the measured core. Solving $t = t_{\max}$ for the depth yields

$$L_{\max} \approx 1 + \frac{\log_{10} t_{\max} - \log_{10} K - \log_{10} \tau(L)}{\log_{10} W}, \quad (\text{A.51})$$

which closes with one or two fixed-point iterations because τ depends on L . Repeated over widths and over the four attackers, it traces the frontiers of Figure 5.12a. On a different machine only τ changes, and a factor s in τ shifts every frontier by $-\log_{10} s / \log_{10} W$, i.e. by 0.19 layers at $W = 40$ for a factor of two.

STEP 6. THE RUNNING TIME OF THE DP. In Figures 5.12b and 5.12c, the enumeration curves are $K W^{L-1} \tau(L)$ of step 3, and the DP curves are the measured running time of the DP implementation on the same core, fitted to the form of its cost, $O(L W^3 G + K W)$. Against the depth at $W = 4$ the fit is affine, $44 \text{ ms} + 0.85 \text{ ms} \times L$, measured for $L = 3$ to 32. Against the width at $L = 3$ it is a cubic polynomial in W , measured up to $W = 96$ and extrapolated beyond, whose leading terms are $10.6 \mu\text{s} \times W^3$ for the propagation and $11.5 \text{ ms} \times W$ for the lookups.